# Understanding Customer Churn Across Satisfaction Segments: An Explainable Machine Learning Approach for CRM Retention Strategy

Student Number: 259366102

Name: Chin Ju Lin (Wendy)

Supervisor: Milad Keshvari Fard

MSc Business Analytics

Submitted as part of the requirement for completing an MSc in Business Analytics at the University of Bath

2 September 2026

Word count: 14,983

# Abstract

Customer retention is a central concern in e-commerce, where relationships are non-contractual and churn is not directly observed. Satisfaction is widely treated as a determinant of retention, yet satisfied customers still leave and dissatisfied customers often stay. Existing churn models typically include satisfaction as one predictor among many, producing a single averaged explanation that may conceal differences between customer groups. This dissertation instead treats satisfaction as a stratifying variable and examines whether the characteristics associated with recorded churn differ systematically across satisfaction levels.

The study analyses a secondary e-commerce dataset of 5,630 customer records, of which 948 (16.8%) are recorded as churned. Customers were divided into low, medium and high satisfaction segments at breakpoints identified in the churn rate distribution. A pooled XGBoost model and three segment-specific models were estimated on a common predictor set and evaluated using stratified five-fold cross-validation. SHAP was applied to compare feature importance and direction within each segment, bootstrap resampling assessed the stability of those rankings, and RFM-informed profiles provided behavioural context. Five robustness diagnostics were conducted before interpretation: label permutation, preprocessing configuration, feature ablation, learning curves and duplicate records.

All four models separated churned from retained customers with ROC-AUC above 0.97 and Average Precision between 0.8895 and 0.9849, against prevalence baselines of 0.119 to 0.238. No diagnostic identified a mechanical source of inflation. However, because the outcome represents a customer status assigned in the dataset rather than an independently observed act of defection, and the timing of that assignment is undocumented, these results demonstrate separability within the dataset rather than verified prospective accuracy. Customer tenure ranked first in every model, with a consistent direction and complete bootstrap stability. Below customer tenure, the composition diverged: cashback amount ranked second in the low satisfaction model, complaint status ranked second in the medium satisfaction model, and registered addresses ranked second in the high satisfaction model,

where delivery distance was also prominent. Directions were largely consistent across segments; the differences lay in relative prominence. RFM profiling showed that recent ordering combined with low frequency and low monetary value carried the highest recorded churn rate within every segment.

The findings support compositional heterogeneity: satisfaction segments share a stable predictive core but distribute information differently across the same set of features, rather than being governed by separate churn mechanisms. For CRM practice, this implies combining common lifecycle priorities with segment-sensitive diagnosis, directing monetary-value monitoring towards low satisfaction customers, complaint resolution towards the medium satisfaction group, and account and delivery review towards high satisfaction customers. These recommendations are presented as testable managerial hypotheses that require prospective validation.

# Table of Contents


Chapter 1: Introduction ..... 1

1.1 Research Background and Rationale ..... 1

1.2 Research Aim and Research Questions ..... 2

1.3 Research Approach ..... 3

1.4 Summary of Key Findings ..... 3

1.5 Dissertation Structure ..... 4

Chapter 2: Literature Review ..... 5

2.1 E-Commerce Customer Churn ..... 5

2.2 Customer Satisfaction and Loyalty ..... 6

2.3 Satisfaction as a Stratifying Variable: Rationale and Justification ..... 7

2.4 Explainable Machine Learning for Churn Prediction (XGBoost, SHAP) ..... 9

2.5 RFM-Based Behavioural Profiling for Interpreting Recorded Churn ..... 10

2.6 Research Gap and Positioning of the Present Study ..... 11

Chapter 3: Methodology ..... 13

3.1 Research Design ..... 13

3.2 Data Source and Collection ..... 13

3.3 Operationalisation of Key Constructs ..... 14

3.3.1 Operationalisation of Churn ..... 14

3.3.2 Satisfaction Group Construction ..... 15

3.4 Analytical Methods ..... 15

3.4.1 Data Preparation ..... 16

3.4.2 Exploratory and Descriptive Analysis ..... 16

3.4.3 Predictive Modelling ..... 17

3.4.4 Model Interpretation ..... 18

3.4.5 Behavioural Profiling ..... 18

3.5 Software ..... 19

3.6 Limitations of Methodology ..... 19

3.7 Ethical Considerations ..... 20

Chapter 4: Data Analysis ..... 21
4.1 Dataset Description ..... 21
4.2 Data Preprocessing ..... 23
4.2.1 Data Cleaning ..... 23
4.2.2 Missing Value Treatment ..... 23
4.2.3 Categorical Encoding ..... 24
4.3 Exploratory Data Analysis ..... 25
4.3.1 Distribution of Variables ..... 25
4.3.2 Churn Rate by Satisfaction Score ..... 28
4.3.3 Behavioural Differences by Churn Status ..... 30
4.3.4 Correlation Analysis ..... 32
4.4 XGBoost-Based Churn Modelling Across Satisfaction Groups ..... 35
4.4.1 Modelling Rationale and Specification ..... 35
4.4.2 Cross-Validated Model Performance ..... 36
4.4.3 Robustness Checks and Performance Diagnostics ..... 37
4.4.4 Preliminary Feature Importance from XGBoost ..... 39
4.5 SHAP-Based Interpretation of Churn Drivers Across Satisfaction Groups ..... 42
4.5.1 Purpose and Interpretation of SHAP Analysis ..... 42
4.5.2 SHAP Feature Importance and Direction Within Satisfaction Segments ..... 42
4.5.3 Bootstrap Stability of SHAP Feature Rankings ..... 46
4.6 RFM-Informed Behavioural Profiling ..... 49
4.6.1 Profile Construction ..... 49
4.6.2 RFM Dimension-Level Churn Patterns ..... 50
4.6.3 Cross-Segment Comparison of Combined RFM Profiles ..... 53
Chapter 5: Discussion and Managerial Implications ..... 56
5.1 Chapter Introduction ..... 56
5.2 Discussion of the Main Findings ..... 56
5.2.1 Descriptive Patterns Within Satisfaction Segments ..... 56
5.2.2 Predictive Performance and Interpretive Boundaries ..... 58
5.2.3 Integrating SHAP and RFM-Informed Behavioural Insights ..... 59

5.3 Managerial Implications and CRM Application ..... 61
5.3.1 From Prediction to CRM Decision-Making ..... 61
5.3.2 Shared Retention Priorities ..... 61
5.3.3 Segment-Differentiated Retention Strategies ..... 62
Chapter 6: Conclusions ..... 64
6.1 Chapter Introduction ..... 64
6.2 Key Conclusions ..... 64
6.3 Contributions and Managerial Relevance ..... 65
6.4 Limitations and Future Research ..... 66
6.5 Final Statement ..... 66
References ..... 67

# List of Tables

Table 4.1: Description of Variables....22
Table 4.2: Recorded Churn Rate by Satisfaction Score ....28
Table 4.3: Satisfaction Groups....28
Table 4.4: Mean Differences Between Churned and Retained Customers by Satisfaction Group ....30
Table 4.5: Cross-Validated Performance of the Pooled and Satisfaction-Specific XGBoost Models ....36
Table 4.6: Summary of Robustness Diagnostics and the Unverifiable Temporal Ordering of the Outcome ....39
Table 4.7: SHAP Feature Rankings and Predominant Directions Across the Pooled and Satisfaction-Specific Models ....46
Table 4.8: Bootstrap Stability of SHAP Feature Rankings (50 Resamples, Eight Highest-Ranked Features per Segment)....48
Table 4.9: Definition and Operational Ranges of the RFM Profile Bands ....50
Table 4.10: Recorded Churn Rate of R3_F2_M1 Relative to Satisfaction-Segment Baselines.55

# List of Figures

Figure 4.1: Distribution of Categorical Variables ..... 26
Figure 4.2: Distribution of Numeric Variables ..... 27
Figure 4.3: Boxplots of Numeric Variables ..... 28
Figure 4.4: Recorded Churn Rate by Original Satisfaction Score ..... 29
Figure 4.5: Recorded Churn Rate Across the Range of Four Selected Variables ..... 32
Figure 4.6: Spearman Correlations with Recorded Churn Across the Pooled Sample and Satisfaction Groups ..... 33
Figure 4.7: Spearman Correlations Among the Predictors ..... 34
Figure 4.8: Learning Curve for the Pooled XGBoost Model ..... 38
Figure 4.9: Gain-Based XGBoost Feature Importance Across Satisfaction Groups ..... 41
Figure 4.10: SHAP Summary Plot for the Low-Satisfaction Model ..... 43
Figure 4.11: SHAP Summary Plot for the Medium-Satisfaction Model ..... 43
Figure 4.12: SHAP Summary Plot for the High-Satisfaction Model ..... 44
Figure 4.13: Recorded Churn Rate by Recency Band and Satisfaction Segment ..... 50
Figure 4.14: Recorded Churn Rate by Frequency Band and Satisfaction Segment ..... 51
Figure 4.15: Recorded Churn Rate by Monetary Band and Satisfaction Segment ..... 52
Figure 4.16: Recorded Churn Rate by Combined RFM Profile and Satisfaction Segment ..... 54

# Chapter 1: Introduction

## 1.1 Research Background and Rationale

In the highly competitive e-commerce environment, customer retention has a significant impact on a company's recurring revenue. Unlike subscription-based services or industries with clear contractual relationships, most e-commerce transactions involve non-contractual relationships. Companies cannot be certain whether customers will make repeat purchases, and it is also difficult to clearly determine when a customer relationship has truly ended. Therefore, retaining existing customers and reducing customer churn have become critical CRM challenges for e-commerce companies. Especially given that acquiring new customers requires ongoing investment in marketing resources, increasing the retention rate of existing customers may be a more efficient path to growth.

Customer satisfaction is typically viewed as a key factor influencing loyalty and retention; higher satisfaction is generally expected to lead to a greater willingness to repurchase and a lower risk of churn. However, satisfaction does not always correlate perfectly with actual retention behavior. Satisfied customers may still stop purchasing, while lower satisfaction does not necessarily lead to churn. One reason for this is that satisfaction primarily reflects customers' evaluations of their past purchasing experiences; this metric alone is insufficient to fully explain whether they will continue to make purchases in the future. Therefore, to understand customer churn, it is necessary to examine not only satisfaction but also customers' behavioral, transactional, and relationship characteristics.

The heterogeneity among customers further complicates the relationship between satisfaction and retention. Even with similar levels of satisfaction, different customers may still exhibit different retention outcomes; therefore, analyzing all customers as a single, homogeneous group may obscure meaningful churn patterns across different segments. Satisfaction thus provides a theoretically sound and intuitive basis for segmentation, enabling research to compare whether customers with different levels of satisfaction exhibit differences in churn-related characteristics.

In particular, high and low levels of satisfaction represent relatively different customer evaluations, yet customers in both groups may ultimately be recorded as churned. This suggests that customers across different satisfaction groups may exhibit different churn-related signals, and that the same behavioural characteristics may vary in importance under different satisfaction conditions. Existing churn models commonly incorporate satisfaction as one of several predictors, leaving less understanding of whether churn-related characteristics vary systematically across satisfaction levels. These considerations motivate the present study to examine whether the characteristics associated with customer churn differ across levels of customer satisfaction.

## 1.2 Research Aim and Research Questions

Accordingly, this study uses customer satisfaction as a stratification variable to compare the behavioural and transactional characteristics of churned and retained customers across satisfaction segments, and to examine how these characteristics are associated with recorded churn. Churn is modelled using machine learning, and interpretability is strengthened through SHAP analysis and RFM-informed behavioural profiling. By identifying shared and segment-specific churn signals, this study aims to provide firms with more targeted CRM retention strategies.

The aim is pursued through four research questions:

RQ1. What descriptive differences in customer characteristics are observed between customers recorded as churned and retained within each satisfaction segment?

RQ2. How effectively do the pooled and satisfaction-specific XGBoost models distinguish between customers recorded as churned and retained, and how robust are their performance estimates?

RQ3. Which predictors are most influential within each satisfaction-specific model, and how do their SHAP importance, predominant direction and ranking stability differ across satisfaction segments?

RQ4. What additional behavioural context do RFM-informed profiles provide across satisfaction segments, and how can the combined findings inform differentiated CRM retention strategies?

## 1.3 Research Approach

This study adopts a quantitative analytical approach using a customer-level e-commerce dataset. Customers are first stratified into different satisfaction groups to allow churn-related patterns to be examined under different satisfaction conditions. Descriptive analysis is used to compare the behavioural and transactional characteristics of churned and retained customers within and across these groups.

Separate XGBoost classification models are then developed for each satisfaction segment to assess whether customer churn can be distinguished using the available customer characteristics. SHAP analysis is subsequently applied to interpret the fitted models and identify both shared and segment-specific characteristics associated with churn. Rather than focusing solely on predictive performance, the analysis therefore emphasises how the relative importance and direction of churn-related characteristics vary across satisfaction groups.

Finally, RFM-based behavioural profiles are used to provide additional context for the identified churn patterns. The findings are then integrated to derive differentiated CRM retention implications for customers with different levels of satisfaction. Full details of the data preparation, analytical procedures and model evaluation are provided in Chapter 3.

## 1.4 Summary of Key Findings

This study defines churn as the customer status recorded at the status assigned and recorded by the firm, rather than as an inactivity event inferred from behaviour. This definition explains two counterintuitive findings: churned customers recorded shorter intervals since their last order, and the churn rate increased as satisfaction rose.

Overall, the three satisfaction segments exhibited both a common core and segment-specific differences. Customer tenure was the most influential predictive feature across all three

segments, with consistent directionality; the differences lay in the relative prominence and combination of the remaining features. Cashback received and product category were more prominent in the low satisfaction segment, complaint status in the medium satisfaction segment, and registered addresses and delivery distance in the high satisfaction segment. The three segments therefore do not have different churn factors, but rather show a different distribution of information across the same set of features.

RFM behavioural profiling further shows that the combination of recent ordering, low frequency and low monetary value corresponded to the highest churn rate within each satisfaction segment.

The pooled model and the low, medium and high satisfaction models all demonstrated high discriminatory power and passed multiple robustness checks. This indicates that the data contain a strong predictive structure, but it should not be interpreted as the predictive accuracy the models would achieve in a deployment setting.

These results support the design of differentiated retention strategies based on satisfaction segmentation, although the resulting recommendations should be treated as testable managerial hypotheses rather than established causal interventions.

## 1.5 Dissertation Structure

Chapter 2 reviews the relevant literature, covering the definition and business significance of e-commerce customer churn, heterogeneity in the relationship between satisfaction and customer outcomes, the rationale for using satisfaction as a stratifying variable, and the application of explainable machine learning and RFM-informed behavioural profiling in churn research. The chapter concludes by identifying the gap in the existing literature and positioning the present study in relation to it.

Chapter 3 sets out the research methodology. It describes the quantitative research design and secondary data analysis strategy, the operationalisation of churn and the satisfaction segments, and the analytical methods employed: data preparation, exploratory and descriptive analysis, XGBoost modelling, SHAP interpretation, bootstrap assessment of

ranking stability, and RFM-informed behavioural profiling. The chapter also reports the software used, the limitations of the methodology and the ethical considerations.

Chapter 4 reports the results of the analysis, presenting in turn the dataset description and preprocessing, the exploratory data analysis, the cross-validated performance and robustness diagnostics of the pooled and satisfaction-specific models, the SHAP feature importance and direction, and the recorded churn rates of the RFM profiles. The chapter focuses on presenting the results, with interpretation reserved for Chapter 5.

Chapter 5 integrates these results and addresses the research questions, distinguishing the churn-related signals shared across the three satisfaction segments from those specific to particular segments, evaluating model performance and defining the boundaries within which the findings can be interpreted. It then translates these findings into managerial implications for CRM retention practice and develops differentiated retention recommendations for the low, medium and high satisfaction segments.

Chapter 6 summarises the main conclusions of the study, sets out its theoretical and practical contributions, and discusses the limitations of the research together with directions for future work.

# Chapter 2: Literature Review

## 2.1 E-Commerce Customer Churn

Customer churn is the counterpart of customer retention: where retention is the customer continuing to transact with the firm, churn is the cessation of that activity (Ascarza et al., 2018). Retaining customers has become a leading priority for firms operating under intense competition (Tamaddoni et al., 2016), and particularly so in online retail, where competing platforms offer overlapping product ranges and switching costs for the customer are low (Matuszelański and Kopczewska, 2022).

Defining the point of churn is considerably harder here than in contractual industries. In contractual settings the firm directly observes the moment a customer terminates the relationship, whereas in non-contractual settings no formal declaration of termination is made.

Online retail belongs to the latter: customers may stop purchasing without notifying the firm, so churn is not an observed event but a latent state, termed latent attrition in the customer-base analysis literature (Ascarza et al., 2018). Whether a customer has churned must therefore be inferred or recorded by other means.

Two approaches follow. The first infers churn from behaviour, most simply by treating a customer as churned after a defined period of inactivity. The window is set by the analyst and varies widely: reported thresholds run from one month to one year depending on the service and the frequency of customer interaction (Jahan and Sanam, 2026), and studies of online retail have variously defined churn as three months without a purchase, a below-average purchase frequency, or the absence of any repeat purchase (Matuszelański and Kopczewska, 2022). The resulting churn rate is therefore in part an artefact of that choice.

The second approach uses the customer status recorded in the firm's own system. This avoids the arbitrariness of a chosen window, but the status reflects the firm's own criteria rather than an externally verifiable behavioural event, and those criteria may be administrative, registering a change in account standing, or declarative, registering a customer's stated decision not to purchase again. Where the recording procedure is not documented, as is common in secondary datasets, the two cannot be distinguished. The present study follows the second approach; the label is defined and its implications set out in Section 3.3.1.

Ascarza et al. (2018) further note that many online settings are hybrid, in that customers may either formally close an account or simply cease to transact. A further complication is specific to e-commerce: customers commonly hold accounts with several competing platforms at once. Reduced purchasing, or a stated intention not to continue purchasing, may therefore represent a shift in share of wallet rather than the end of the relationship.

## 2.2 Customer Satisfaction and Loyalty

Customer satisfaction is generally associated with stronger loyalty and lower churn, but the relationship is neither complete nor uniform. Kumar, Dalla Pozza and Ganesh (2013) find that satisfaction alone explains only a small proportion of the variation in loyalty and retention

behaviour, while Oliver (1999) argues that satisfaction is a necessary but insufficient condition for loyalty, as many customers defect despite reporting positive evaluations of the firm.

One reason lies in the structure of loyalty itself. Dick and Basu (1994) distinguish attitudinal loyalty, reflecting psychological commitment to a brand, from behavioural loyalty, reflected in actual repurchase. Because the two do not always move together, customers reporting similar satisfaction may nonetheless differ in whether they remain.

The strength of the relationship also varies across customers. A latent segmentation analysis identifies three segments in which the satisfaction–loyalty relationship differs significantly, and shows that analysing customers as a single homogeneous population overstates the overall effect of satisfaction on loyalty (Fuentes-Blasco, Moliner-Velázquez and Gil-Saura, 2014). Heterogeneity has likewise been argued to require explicit consideration (Human and Naude, 2014). Satisfaction is therefore better treated as one indicator of churn risk than as a sufficient explanation of it.

## 2.3 Satisfaction as a Stratifying Variable: Rationale and Justification

Given the evidence that satisfaction-related outcomes may vary across customer groups, a key methodological issue is how customer satisfaction should be incorporated into churn analysis.

In many churn prediction studies, satisfaction is treated as a predictor variable and entered directly into the model alongside behavioural, demographic and transactional features (Boukrouh and Azmani, 2025; Jahan and Sanam, 2026). This approach is useful when the aim is to estimate the average association between satisfaction and churn. However, it is less suitable when the research aim is to examine whether different satisfaction groups may be associated with different churn mechanisms.

Market segmentation divides a heterogeneous customer population into groups that are internally more homogeneous, on the expectation that members of a segment will respond similarly to marketing effort (Cooil, Aksoy and Keiningham, 2007). This logic is central to CRM because retention strategy depends not only on identifying customers at risk, but on how different groups should be served. Satisfaction is a defensible basis for such stratification here,

being directly recorded in the dataset and theoretically linked to retention; the breakpoints are set out in Section 3.3.2.

Empirical research also supports the use of satisfaction-related information for segmentation. Satisfaction cues have been shown to support market segmentation and help explain switching behaviour (Athanassopoulos, 2000), which suggests that satisfaction measures can serve not only as explanatory variables but also as a basis for structuring customer analysis. The contribution of product or service attributes to satisfaction has likewise been argued to require examination at the segment level, since the same attribute may play different roles across customer groups (Füller and Matzler, 2008). Aggregate analysis may therefore conceal important segment-specific differences.

The modelling literature provides further support. Assuming a single homogeneous population is often unrealistic when analysing cause-and-effect relationships (Ringle, Sarstedt and Mooi, 2010). Kim, Fong and DeSarbo (2012) similarly show that different customer segments may require different sets of explanatory variables. These studies suggest that customer segments may differ not only in the strength of relationships, but also in the composition of the factors that explain an outcome.

This distinction is important for the present study. The previous section discussed relational heterogeneity, where the strength of the satisfaction–outcome relationship differs across customer groups. This study extends that idea by examining compositional heterogeneity, where the factors associated with churn may differ across satisfaction segments. In this sense, satisfaction-based stratification is not used merely to compare churn rates across groups, but to provide a structure for investigating whether different satisfaction segments are characterised by different behavioural and transactional churn drivers.

This rationale is also consistent with recent churn analytics and CRM research. De Caigny et al. (2021) argue that segmented prediction approaches can provide more granular, segment-specific insights into churn drivers, and segmentation should moreover support actionable customer insight rather than merely classify customers descriptively (Bailey et al., 2009). Treating satisfaction as a stratifying variable therefore provides a literature-supported foundation for generating segment-specific churn insights that can inform CRM retention

strategy. The operational details of satisfaction segmentation, model development and explainability analysis are discussed in the methodology chapter.

## 2.4 Explainable Machine Learning for Churn Prediction (XGBoost, SHAP)

As firms gain access to increasingly rich customer data, customer churn prediction has gradually moved beyond traditional statistical models towards machine learning and ensemble learning approaches (Ahmad, Jafar and Aljoumaa, 2019). Traditional models such as logistic regression remain valued for their interpretability, but customer churn is usually influenced by multiple behavioural, transactional and demographic factors (Maulana and Hidayati, 2025) that may interact in non-linear ways, and such relationships must be specified manually in a linear model. For this reason, tree-based ensemble models, such as random forest and XGBoost, have become increasingly common in churn prediction research because they can capture more complex patterns in customer behaviour (Ahmad, Jafar and Aljoumaa, 2019; Maulana and Hidayati, 2025).

Among these methods, XGBoost is an ensemble learning algorithm based on gradient boosting decision trees (Al-Shatnwai and Faris, 2020). Its objective includes a regularisation term that penalises model complexity and so limits overfitting (Chen and Guestrin, 2016). It has been widely applied in customer churn prediction studies (Al-Shatnwai and Faris, 2020), including in e-commerce (Gan, 2022).

However, predictive performance alone does not guarantee managerial usefulness. Chang et al. (2024) emphasise that, for CRM retention strategy, firms need to know not only which customers are likely to churn, but also why they are predicted to be at risk. If a model only produces churn probabilities without explaining the factors that drive those predictions, managers may struggle to translate model outputs into concrete retention actions.

Explainable artificial intelligence (XAI) provides important tools for addressing this issue. SHAP (SHapley Additive exPlanations) is one widely used model explanation method (Chang et al., 2024). It can provide global explanations, which identify the most important churn drivers across the overall model, as well as local explanations, which explain why an individual

customer is predicted to have a high churn risk. This dual explanatory capacity makes SHAP particularly useful in CRM contexts, as it helps translate model outputs into more commercially meaningful customer insights.

For the present study, the value of SHAP lies not only in explaining the overall model, but also in comparing churn drivers across satisfaction segments. As Section 2.3 argued, aggregate analysis may conceal segment-specific differences; where it does, a global explanation describes the average customer rather than any particular group. De Caigny et al. (2021) extend this reasoning to interpretation, presenting model explanations at the segment level rather than only globally. Applying SHAP within each satisfaction segment follows the same logic and can help identify which behavioural and transactional factors are most influential for each group.

This aligns with the study's focus on compositional heterogeneity: different satisfaction groups may not simply be affected by the same factors to different degrees, but may instead be driven by different combinations of churn-related factors. Applications of SHAP in churn research, including those discussed above, have nonetheless concentrated on explaining a single model estimated over the full customer base. This distinction between degree and composition is what the segment-level application of SHAP is intended to capture.

## 2.5 RFM-Based Behavioural Profiling for Interpreting Recorded Churn

RFM analysis provides a widely used behavioural and transactional framework for summarising customer engagement (Brahmana, Mohammed and Chairuang, 2020; Aslantaş et al., 2023). It captures three dimensions of customer behaviour: recency, how recently a customer has purchased; frequency, how often a customer purchases; and monetary value, the customer's historical spending level (Ullah et al., 2023). Together these provide an interpretable way to profile customer activity and historical transactional value.

RFM variables are widely applied in churn analysis because they link customer outcomes to observable patterns of transactional behaviour. In much of this work, however, churn risk is inferred from the RFM profile itself rather than measured against an observed outcome. Zaghloul, Barakat and Rezk (2025), for instance, apply K-means clustering to RFM scores in

order to generate churn labels for datasets in which no churn outcome is recorded, so that the label reflects a customer's position on the RFM dimensions rather than any observed churn event. Wilbert et al. (2023) similarly designate one of the segments obtained from RFM clustering as lost customers on the basis of their position across the three dimensions. Such designs describe transactional structure, but cannot establish whether the customers so identified in fact churn, since no outcome is observed against which the profile could be assessed.

The present study therefore separates the two roles. Churn is operationalised independently of RFM, and RFM serves as a behavioural profiling framework for examining whether customers with different recorded churn outcomes display different transactional patterns.

This separation also matters for how satisfaction is used. Satisfaction and RFM capture different kinds of evidence. Satisfaction is attitudinal and evaluative, whereas RFM is behavioural and transactional, and customers reporting the same level of satisfaction may still differ substantially in purchase recency, purchase frequency and spending level. RFM therefore allows churned and retained customers to be compared within a satisfaction segment rather than across segments alone.

Aslantaş et al. (2023) argue that RFM-based segmentation can support more targeted marketing actions by distinguishing customers according to their transactional value and activity patterns. Customers with high historical spending but declining recent activity, for example, may require a different retention approach from customers with low frequency and low monetary value (Zaghloul, Barakat and Rezk, 2025).

## 2.6 Research Gap and Positioning of the Present Study

Despite a substantial body of research on customer satisfaction, customer heterogeneity and machine learning-based churn prediction, these streams of literature remain insufficiently integrated. The satisfaction–loyalty literature has increasingly shown that satisfaction does not translate uniformly into loyalty or retention outcomes, and that the strength of this relationship may vary across customer groups (Grewal, Chandrashekaran and Citrin, 2010; Fuentes-Blasco, Moliner-Velázquez and Gil-Saura, 2014). At the same time, churn prediction

studies have demonstrated the value of machine learning models for identifying customers who are at risk of churn (Tamaddoni, Stakhovych and Ewing, 2016; Zaghloul, Barakat and Rezk, 2025). These two streams have developed largely in parallel: satisfaction is typically entered into churn models as one predictive feature among many, rather than used as a basis for structuring the analysis itself.

Tamaddoni, Stakhovych and Ewing (2016) note that the absence of service-related variables, including customer satisfaction, creates opportunities for future research to examine these effects. The present study builds on this point by treating satisfaction not as an additional predictor but as a stratifying dimension for investigating whether churn drivers differ across satisfaction-defined groups.

The gap in segment-level explanation identified in Section 2.4 compounds the first. Where a single global explanation is the only output available, it may not provide sufficiently specific guidance for designing segment-level interventions, and the analytical limitation therefore becomes a managerial one.

The present study addresses these gaps by treating satisfaction as a stratifying variable rather than as a predictive feature alone, estimating separate models within each satisfaction segment, and comparing the resulting SHAP explanations across segments. RFM-informed behavioural profiling then provides transactional context for interpreting the recorded churn status.

In doing so, this dissertation positions itself at the intersection of satisfaction heterogeneity, explainable churn prediction and CRM retention strategy. Its contribution is both analytical and managerial. Analytically, it extends work on relational heterogeneity to compositional heterogeneity. Managerially, it seeks to translate these segment-level differences into more differentiated and actionable retention strategies.

# Chapter 3: Methodology

## 3.1 Research Design

This study adopts a quantitative research choice and a secondary data analysis strategy. The research questions concern whether the factors associated with customer churn differ systematically across levels of reported satisfaction; a quantitative design is therefore appropriate. Secondary data are used because a dataset containing both a recorded churn outcome and a satisfaction measure at customer level was available, and collecting comparable primary data would not have been feasible within the scope of this dissertation.

The analytical strategy is stratified rather than pooled. Rather than estimating a single churn model for all customers, the sample is divided into satisfaction segments and a separate model is estimated within each, so that the relative influence of each predictor is allowed to differ between segments. A pooled model is retained as a benchmark. The comparison of results across segments, rather than the predictive performance of any single model, constitutes the analytical focus of the study.

## 3.2 Data Source and Collection

This study uses a publicly available secondary dataset entitled *Ecommerce Customer Churn Analysis and Prediction*, obtained from Kaggle and uploaded by Ankit Verma (Verma, 2021). It was accessed in Microsoft Excel format in June 2026 and selected because it records customer-level churn status alongside satisfaction and a range of behavioural, transactional and profile characteristics.

The dataset contains 5,630 observations and 20 variables, each row representing one customer. All records are retained in the analytical sample, with missing values treated as described in Section 3.4.1.

The accompanying documentation provides brief variable descriptions but does not identify the original platform, market, sampling frame, collection procedure, observation period or extraction date, and provides no longitudinal records, churn date or predictor timestamps. The description of the Churn field is similarly limited (Section 3.3.1). The dataset is therefore

treated as a customer-level extract with limited source documentation, the consequences of which are discussed in Section 3.6.

## 3.3 Operationalisation of Key Constructs

Two definitional decisions follow from the properties of the dataset described above: how the churn outcome is interpreted, and how the satisfaction segments are constructed.

### 3.3.1 Operationalisation of Churn

Churn is understood conceptually as the cessation of a customer's relationship with the firm (Section 2.1). Operationally, the dependent variable throughout this study is the binary Churn indicator supplied with the source dataset, which records each customer's status. The data dictionary describes the field only as a "Churn Flag" and does not document how the status was determined or when it was recorded.

Three features of the data indicate that it was not derived from a purchase-inactivity rule of the kind reviewed in Section 2.1. First, DaySinceLastOrder ranges from 0 to 46 days and cannot represent the inactivity periods of one month to one year on which such definitions are conventionally based. Second, the indicator contradicts an inactivity reading in both directions: of the 894 churned customers with a recorded value, 474 (53.0%) had ordered within the previous two days, while 707 of the 4,429 retained customers (16.0%) had not ordered for at least nine days. Third, no threshold on DaySinceLastOrder reproduces the indicator, the best-performing rule merely matching the majority-class baseline of 0.832 while assigning almost no customer to the churned class.

The indicator is therefore treated as a recorded churn status rather than as a behavioural label reconstructed from transaction history. Because the recording procedure is undocumented, the status may represent an administrative classification or a customer's stated intention not to purchase again. The data do not distinguish between these possibilities and neither is asserted here.

This definition has one direct consequence for model specification. Variables that would conventionally be used to construct an inactivity-based outcome, particularly DaySinceLastOrder, are not required for that purpose and are retained as explanatory features.

Their relationships with the outcome may therefore run counter to the direction expected under an inactivity definition, as reported in Sections 4.3.3 and 4.6.2. The interpretive limits arising from the undocumented recording procedure are set out in Section 3.6.

### 3.3.2 Satisfaction Group Construction

For segment-specific analysis, the five-level SatisfactionScore variable is collapsed into three groups. Retaining all five levels would leave the smallest category with 586 customers and 74 churn cases, which is insufficient for stable segment-specific estimation.

The breakpoints are selected by inspecting the recorded churn rate across the 1 to 5 scale, reported in Section 4.3.2. Churn rates within each resulting pair differ by no more than 1.1 percentage points, whereas the two selected breakpoints correspond to the largest gaps in the distribution, of 4.6 percentage points between scores 2 and 3 and 6.7 points between scores 4 and 5. The grouping also remains interpretable on the ordinal scale, distinguishing the dissatisfied end, a neutral-to-moderate level and full satisfaction, although the separation of scores 4 and 5 follows the observed churn pattern rather than the conventional midpoint of the scale.

Because the breakpoints derive from the recorded churn rate, satisfaction group is not treated as an independent predictor of churn, and differences in baseline churn rate between groups are a property of the grouping rule rather than a finding. SatisfactionScore is excluded from the segment-specific models for the same reason. The selection rule used only the marginal churn rate at each score and no information about any predictor, so the comparison of feature importance across groups is not circular; cross-group comparisons are nonetheless interpreted descriptively.

## 3.4 Analytical Methods

The analysis proceeds in five stages: data preparation, exploratory and descriptive analysis, predictive modelling, model interpretation, and behavioural profiling. This section outlines the methods used at each stage and the rationale for selecting them. Implementation details and results are reported in Chapter 4.

### 3.4.1 Data Preparation

Missing values are present in seven variables at rates of between 4.5% and 5.5%. Because a third of records contain at least one missing value, imputation is used rather than listwise deletion. Median imputation is applied to the six numeric variables affected, and mode imputation to the discrete usage variable HourSpendOnApp. Imputation is embedded within the modelling pipeline so that its statistics are learned only from the training portion of each cross-validation fold, preventing information transfer between training and evaluation data; in the descriptive and profiling analyses, which involve no model fitting, it is applied to the dataset as a whole.

One-hot encoding, with the first level dropped to avoid redundancy, is applied only within the modelling pipeline; the descriptive analyses use the original category labels. CustomerID is excluded as a record identifier and SatisfactionScore for the reason given in Section 3.3.2. Variable-level treatment is reported in Section 4.2.

### 3.4.2 Exploratory and Descriptive Analysis

Before predictive modelling, churn rates are calculated across the original satisfaction scores and the three segments defined in Section 3.3.2, which also explains why differences in baseline churn rates between segments are not treated as independent findings.

Within each segment, selected behavioural and transactional variables are compared between customers recorded as churned and retained, with mean differences calculated as the churned-group mean minus the retained-group mean. Spearman correlations with the binary Churn indicator are calculated for the pooled sample and separately within each segment, restricted to numeric and binary variables. Spearman is used because several variables are discrete or non-normally distributed. These analyses are descriptive and are not interpreted as independent or causal effects.

### 3.4.3 Predictive Modelling

Churn is modelled using XGBoost, a gradient-boosted decision tree algorithm (Chen and Guestrin, 2016). It was selected because it accommodates non-linear relationships and variable interactions without requiring them to be specified in advance, performs well on structured tabular data, and is compatible with the SHAP-based interpretation described in Section 3.4.4.

A separate model is estimated for each satisfaction group, alongside a pooled model serving as a benchmark. Segment-specific estimation allows the predictors and their relative importance to vary freely across groups, which is what the study's focus on differences in the composition of churn-related signals requires. All models use the same candidate predictors (Section 3.4.1).

No hyperparameter tuning is undertaken. A common fixed specification is applied to all models, to preserve comparability across analytical samples and to avoid using the same cross-validation folds for both model selection and performance reporting. Class imbalance is addressed using scale_pos_weight, calculated from the non-churn-to-churn ratio within each analytical sample and held constant across its five folds.

Performance is estimated using stratified five-fold cross-validation with shuffling and a fixed random seed (Kohavi, 1995). Stratification preserves approximately the same churn proportion in each fold, reducing instability caused by the minority class, and results are reported as the mean and standard deviation across folds.

Models are evaluated using ROC-AUC, Average Precision (AP) and F1. ROC-AUC assesses the ranking of churned above retained customers against a fixed chance baseline of 0.5. AP is more informative under class imbalance, but its no-skill baseline approximates the churn prevalence of the sample evaluated, so values are read relative to that baseline rather than compared across segments. F1 is supplementary and threshold-dependent, reported at 0.5.

Cross-validation estimates predictive performance only. Final models are then fitted to the complete pooled sample and to the complete sample within each satisfaction segment, providing the basis for the interpretation described in Section 3.4.4. Five further diagnostics addressing potential mechanical sources of inflated performance are reported in Section 4.4.3.

### 3.4.4 Model Interpretation

Feature importance is assessed using SHAP values, which attribute a model's prediction for each observation to its individual input features on a common additive scale (Lundberg and Lee, 2017). SHAP is preferred to the gain-based importance produced natively by XGBoost because it provides both the magnitude and the direction of each feature's contribution. Values are computed using the tree-specific estimator developed for gradient-boosted models (Lundberg et al., 2020).

SHAP values are calculated for the final pooled and segment-specific models. The pooled results serve as an aggregate reference showing how the predictors are used when heterogeneity is not modelled explicitly, while the comparison across satisfaction-specific models is the primary analytical focus. Because each model is explained on the observations to which it was fitted, the values are in-sample explanations rather than out-of-fold ones.

Because rankings estimated from a single fitted sample may be sensitive to sampling variation, ranking stability is assessed within each segment by bootstrap resampling. Fifty samples of the same size as the segment are drawn with replacement, and samples containing only one outcome class are excluded. For each valid resample the class weight is recalculated from its class distribution, the complete preprocessing and modelling pipeline is refitted, and features are ranked by mean absolute SHAP value on the resampled observations.

Stability is summarised using the mean and standard deviation of each feature's rank across resamples. Rank standard deviations below 1.0 are classified as stable, between 1.0 and 1.5 as moderately stable, and above 1.5 as unstable. These thresholds are author-defined and serve as descriptive aids rather than inferential criteria. Results are reported in Section 4.5.3.

### 3.4.5 Behavioural Profiling

The SHAP results are interpreted alongside a behavioural profiling scheme based on the RFM framework (Anitha and Patil, 2022), which classifies customers according to transaction recency, frequency and monetary value. Because the dataset does not contain customers' actual expenditure or transaction-revenue values, three available variables are used as proxies: DaySinceLastOrder for recency, OrderCount for frequency, and CashbackAmount as the

available indicator of monetary value. The limitations of these measures are discussed in Section 3.6.

Customers are assigned to three bands on each dimension using thresholds derived from the full sample rather than separately within each satisfaction segment. This ensures that the same band represents the same behavioural range across all groups. Terciles are used for the recency and monetary dimensions. The frequency dimension is banded directly according to order count because it is a discrete variable for which quantile-based cutting could assign customers with identical values to different bands. Combined profiles are then formed from the three dimensions and compared across satisfaction groups in Section 4.6.3.

Wilson score 95% confidence intervals are calculated for each profile-level churn proportion to represent the sampling uncertainty surrounding the observed rates. Profiles containing fewer than 30 customers are excluded from detailed interpretation to reduce instability arising from small cell sizes.

## 3.5 Software

All analysis was conducted in Python 3.14.3 using Jupyter Notebook. Data manipulation used pandas 3.0.0 and NumPy 2.4.2; preprocessing, cross-validation and evaluation used scikit-learn 1.8.0; gradient boosting used the xgboost package 3.2.0; model interpretation used the shap package 0.52.0; statistical tests used SciPy 1.17.0; and visualisations were produced with Matplotlib 3.10.8 and Seaborn 0.13.2.

## 3.6 Limitations of Methodology

The principal limitation follows from the outcome variable (Section 3.3.1). If the recorded status reflects a stated intention rather than an action already taken, the label is weaker evidence of eventual behaviour than it appears, since intention measures predict subsequent purchasing imperfectly (Morwitz, Steckel and Gupta, 2007). The outcome is therefore read as a leading indicator of disengagement rather than as confirmed attrition.

A related limitation is temporal, and follows from the documentation of the extract rather than from its structure. The dataset carries no timestamps for either the churn status or the predictors, and no observation period or extraction date is recorded (Section 3.2), so the analysis cannot establish that the predictor values preceded the status. A customer already on the point of leaving may have changed their behaviour before the status was assigned, so some predictors may reflect the consequences of impending churn rather than conditions leading to it. The findings are interpreted as associations of unknown temporal ordering, not as antecedents of churn.

Imputation was used to retain the full analytical sample (Section 3.4.1). This affects the descriptive and profiling analyses more than the models, since imputed values concentrate at the relevant median or mode and may compress the RFM bands.

The RFM dimensions are proxies rather than complete measures: CashbackAmount represents a reward received rather than expenditure, and OrderCount has no documented time period, so it reflects accumulated activity rather than a purchase rate. The satisfaction groups are similarly constrained by the grouping rule (Section 3.3.2). SHAP values describe how the fitted models used the predictors and do not demonstrate causal effects. Their sensitivity to sampling variation is assessed through bootstrap resampling (Section 3.4.4).

Finally, the findings derive from a single e-commerce dataset with an unspecified market, period and customer population. They should be read as evidence of differences in churn-related predictive signals within this dataset, rather than as results generalisable to other e-commerce platforms.

## 3.7 Ethical Considerations

This study uses secondary data only. No primary data collection was undertaken, so no participant recruitment, informed consent or non-disclosure agreement was required. The dataset is publicly available and was obtained and used in accordance with the terms under which it is published (Section 3.2).

The dataset contains no direct identifiers such as names, contact details, addresses or payment information, and no free-text fields. The only identifier present is a sequential

customer reference number, which carries no information linking a record to an individual and was excluded from all analyses. The remaining variables describe account and transaction characteristics at a level that does not permit re-identification, and no attempt was made to link the data to any external source. The data therefore fall outside the scope of personal data as defined under the UK General Data Protection Regulation.

The research was reviewed under the University of Bath School of Management ethics procedures.

# Chapter 4: Data Analysis

## 4.1 Dataset Description

The dataset introduced in Section 3.2 consists of 5,630 records and 20 variables, with each record representing one customer. Its composition is described here before the preprocessing procedures reported in Section 4.2.

The variables fall into four analytical groups. CustomerID serves only as a record identifier and is excluded from the analysis. Churn is the binary outcome variable. SatisfactionScore is the ordinal rating used to define the analytical segments, and is excluded from the predictor set for that reason. The remaining 17 variables are retained as candidate predictors, describing customer profile, platform engagement and transactional behaviour. Of these, five are categorical and twelve are numeric, count or binary variables.

Recorded churn is imbalanced, with 948 customers (16.8%) recorded as churned and 4,682 (83.2%) as retained. This imbalance is addressed through class weighting and evaluated using metrics appropriate to skewed classes, as described in Section 3.4.3.

*Table 4.1: Description of Variables*

| Variable | Type | Description | Range / Categories |
|---|---|---|---|
| **Identifier** | | | |
| CustomerID | Integer | Unique record identifier assigned to each customer | 50001-55630 |
| **Outcome** | | | |
| Churn | Binary | Whether the customer is recorded as churned in the source dataset | 0 = retained (4,682); 1 = churned (948; 16.8%) |
| **Stratification** | | | |
| SatisfactionScore | Ordinal | Customer's reported satisfaction rating | 1-5 |
| **Customer profile** | | | |
| Gender | Categorical | Recorded gender of the customer | Male (3,384); Female (2,246) |
| MaritalStatus | Categorical | Recorded marital status | Married (2,986); Single (1,796); Divorced (848) |
| CityTier | Ordinal | Tier of the customer's city, where tier 1 denotes the most developed | 1 (3,666); 2 (242); 3 (1,722) |
| NumberOfAddress | Count | Number of addresses registered to the customer account | 1-22 (median 3) |
| WarehouseToHome | Numeric | Distance from the warehouse to the customer's address, in the units recorded in the source dataset | 5-127 (median 14) |
| **Platform engagement** | | | |
| Tenure | Numeric | Length of the customer's relationship with the platform | 0-61 (median 9) |
| PreferredLoginDevice | Categorical | Device most frequently used to access the platform | Mobile Phone (3,996); Computer (1,634) |
| NumberOfDeviceRegistered | Count | Number of devices registered to the customer account | 1-6 (median 4) |
| HourSpendOnApp | Count | Hours spent on the application or website | 0-5 (median 3) |
| Complain | Binary | Whether a complaint was recorded for the customer | 0 = no complaint (4,026); 1 = complaint recorded (1,604; 28.5%) |
| **Transactional** | | | |
| PreferedOrderCat | Categorical | Product category the customer ordered from most frequently | Mobile Phone (2,080); Laptop & Accessory (2,050); Fashion (826); Grocery (410); Others (264) |
| PreferredPaymentMode | Categorical | Payment method most frequently used | Debit Card (2,314); Credit Card (1,774); E-wallet (614); Cash on Delivery (514); UPI (414) |
| OrderCount | Count | Number of orders recorded for the customer | 1-16 (median 2) |
| DaySinceLastOrder | Count | Number of days since the customer's most recent recorded order | 0-46 (median 3) |
| CouponUsed | Count | Number of coupons redeemed by the customer | 0-16 (median 1) |
| OrderAmountHikeFromlastYear | Numeric | Percentage increase in order value relative to the previous year, as recorded in the source dataset | 11-26 (median 15) |
| CashbackAmount | Numeric | Cashback received by the customer, in the units recorded in the source dataset | 0.00-324.99 (median 163.28) |

## 4.2 Data Preprocessing

Preprocessing involved three steps: standardising category labels, imputing missing values, and encoding categorical variables. The first two apply to all analyses, while encoding is required only for the predictive models, since the descriptive analyses work directly with the original category labels.

Steps that address model-specific requirements were not needed. Numeric variables were not scaled and outliers were not transformed, as gradient-boosted trees split on rank order rather than absolute magnitude. No derived variables were constructed, and no feature selection was carried out; the rationale for retaining a common predictor set is given in Section 4.4.1.

### 4.2.1 Data Cleaning

Three categorical variables used more than one label for the same category. In PreferredPaymentMode, CC and COD were rewritten as Credit Card and Cash on Delivery. In PreferredLoginDevice, Phone was merged into Mobile Phone, and in PreferedOrderCat, Mobile was merged into Mobile Phone. The data dictionary does not distinguish between the labels in each pair, and keeping them apart would split one category across two indicator variables when the data are one-hot encoded.

Some records share the same values on every analytical variable. It cannot be determined whether these are repeated entries for one customer or separate customers with matching attributes, so they were kept. Their effect on the models is examined in Section 4.4.3.

### 4.2.2 Missing Value Treatment

Missing values are identified across seven variables, with column-wise missingness rates ranging from 4.5% to 5.5%. Notably, no customer record contains missing values in more than one variable, indicating that missing values are distributed across distinct columns rather than concentrated within particular records. At the row level, 1,856 records (33%) contain at least one missing value. These missing values were imputed rather than deleted, in order to avoid losing a substantial part of the analytical sample.

Missingness rates differed descriptively between churned and retained customers. Imputation was nevertheless carried out without reference to the Churn indicator, since imputing separately by outcome would introduce information from the outcome into the predictor values. Model-based alternatives such as regression or nearest-neighbour imputation were also avoided, for a different reason: they estimate missing values from the other predictors, and would therefore impose relationships between predictors that are subsequently examined through feature importance analysis.

Median imputation was applied to the six numeric variables with missing values. Several display a long right tail. WarehouseToHome reaches 127 against a third quartile of 20, and CouponUsed and OrderCount both reach 16 against third quartiles of 2 and 3 respectively. Under these distributions the mean would be drawn towards the extreme values, while the median is unaffected. The median also stays within each variable's observed range and preserves its integer scale. Mode imputation was applied to HourSpendOnApp, which takes only six discrete values from 0 to 5 and functions as an ordered category.

For the descriptive analyses, including the RFM-informed profiling, imputation was applied to the dataset as a whole. For predictive modelling, the same procedures were implemented within the modelling pipeline and fitted separately using the training data in each cross-validation fold.

### 4.2.3 Categorical Encoding

Categorical variables were transformed using one-hot encoding, with the first category of each variable dropped to avoid redundancy among the resulting indicators. This produced 24 features from the 17 candidate predictors.

Encoding was applied only to the predictive models in Sections 4.4 and 4.5. The descriptive analyses in Sections 4.3 and 4.6 use the original category labels, since frequency tables and group comparisons operate directly on them. The correlation analysis in Section 4.3.4 is restricted to the numeric and binary variables for the same reason.

Within the modelling pipeline, the encoder was fitted separately on each cross-validation training fold, so that the set of categories was determined from the training data alone.

## 4.3 Exploratory Data Analysis

### 4.3.1 Distribution of Variables

Figures 4.1 to 4.3 show the distribution of the candidate predictors. The categorical variables are unevenly distributed. Debit and credit cards account for the majority of recorded payment methods, mobile phones and laptops and accessories together account for around three-quarters of preferred order categories, and tier 1 cities account for roughly two-thirds of customers. A complaint is recorded for 1,604 customers (28.5%).

Most numeric variables are right-skewed. Tenure, order count, coupons redeemed and days since last order all concentrate at low values with a thinning tail, while NumberOfAddress and WarehouseToHome follow the same shape less steeply. Cashback amount is the exception, forming a single peak around 150 with a moderate right tail. Two records report a WarehouseToHome above 120, against a maximum of 36 elsewhere, and twelve report a cashback amount below 100 against a median of 163.

The variables marked as imputed show a spike at the median or modal value, most visibly at nine months of Tenure, two recorded orders and fourteen for WarehouseToHome.

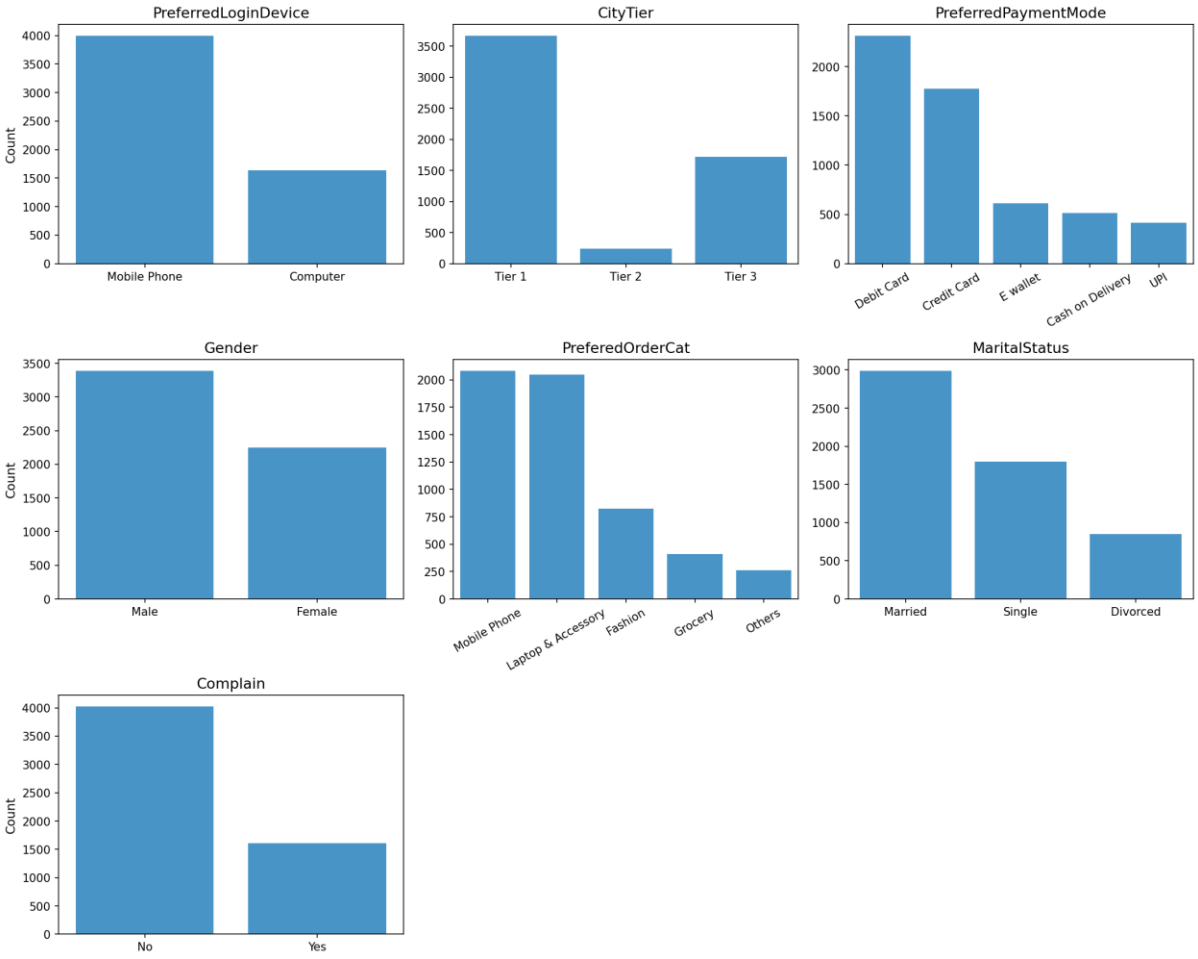



*Figure 4.1: Distribution of Categorical Variables*

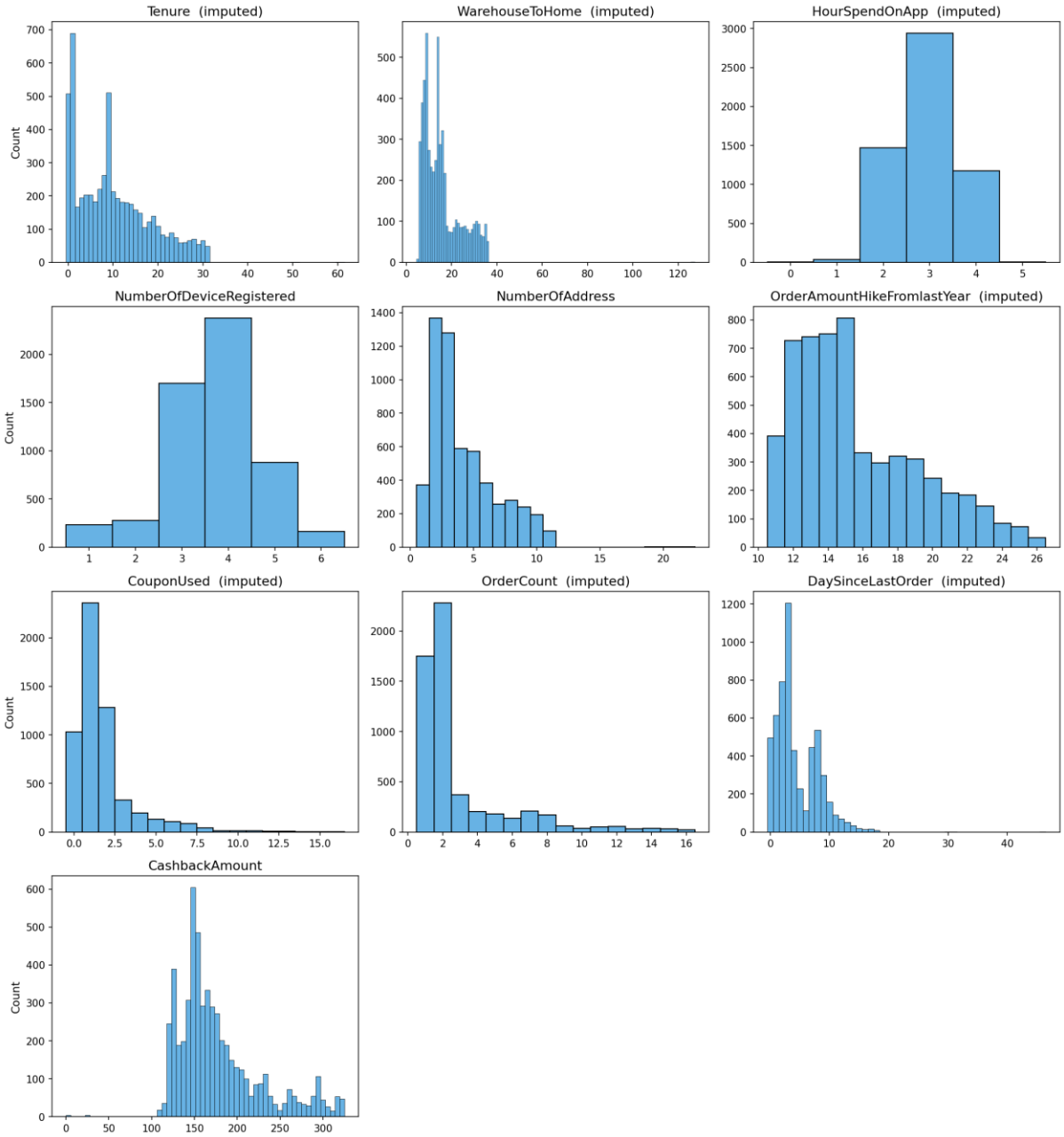



*Figure 4.2: Distribution of Numeric Variables*

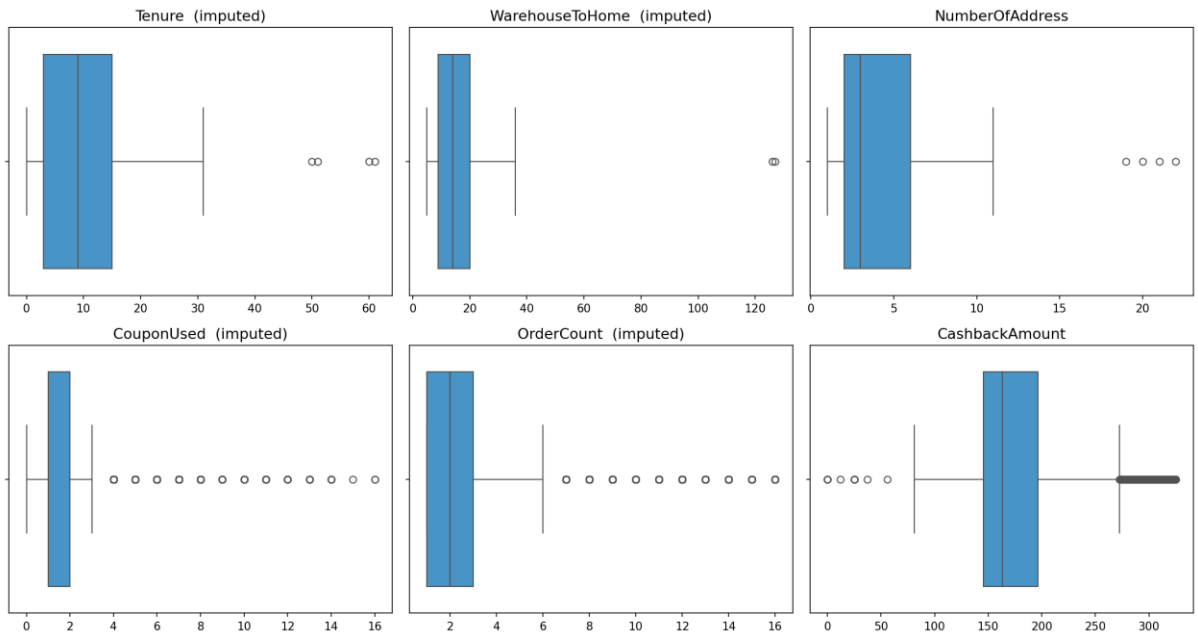



*Figure 4.3: Boxplots of Numeric Variables*

### 4.3.2 Churn Rate by Satisfaction Score

*Table 4.2: Recorded Churn Rate by Satisfaction Score*

| Satisfaction score | Customers | Churned | Churn rate |
|---|---|---|---|
| 1 | 1,164 | 134 | 11.5% |
| 2 | 586 | 74 | 12.6% |
| 3 | 1,698 | 292 | 17.2% |
| 4 | 1,074 | 184 | 17.1% |
| 5 | 1,108 | 264 | 23.8% |

*Table 4.3: Satisfaction Groups*

| Group | Satisfaction scores | Customers | Churned | Churn rate |
|---|---|---|---|---|
| Low | 1–2 | 1,750 | 208 | 11.9% |
| Medium | 3–4 | 2,772 | 476 | 17.2% |
| High | 5 | 1,108 | 264 | 23.8% |
| Total | 1–5 | 5,630 | 948 | 16.8% |

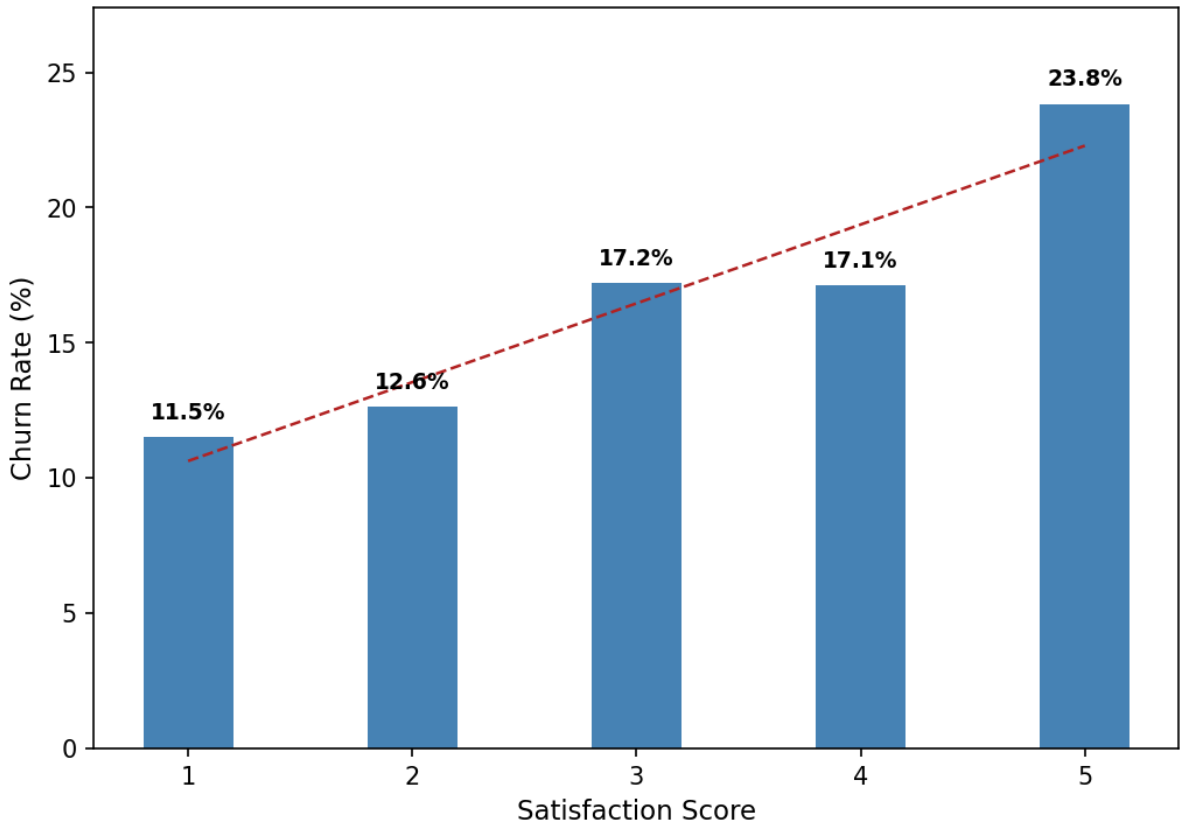



*Figure 4.4: Recorded Churn Rate by Original Satisfaction Score*

Table 4.2 reports the recorded churn rate at each level of the original satisfaction scale. The rate rises from 11.5% at score 1 to 23.8% at score 5. Figure 4.4 shows this pattern together with a fitted linear trend, which has a slope of 2.91 percentage points per scale point.

The increase is concentrated at two points on the scale, between scores 2 and 3 and between scores 4 and 5. These breakpoints define the three satisfaction groups used throughout the remaining analysis, for the reasons set out in Section 3.3.2. Table 4.3 reports the resulting group sizes and churn rates. The high satisfaction group is the smallest, with 1,108 customers, but contains 264 churn cases.

The direction of the relationship runs counter to the conventional expectation that higher satisfaction accompanies stronger retention. Since the group boundaries were derived from this distribution, the difference in baseline churn rate between groups is not treated as an

independent finding. The analyses that follow examine whether behavioural and transactional patterns differ across the groups.

### 4.3.3 Behavioural Differences by Churn Status

Table 4.4 reports the difference in group means for each variable, calculated within each satisfaction group as the mean among churned customers minus the mean among retained customers. A negative value therefore indicates that churned customers recorded a lower average than retained customers. Because the variables are measured on different scales, the magnitudes should be compared across satisfaction groups within a row rather than between rows.

$$\Delta = \bar{x}_{churned} - \bar{x}_{retained}$$

where $\bar{x}$ denotes the group mean of a given variable, calculated separately within each satisfaction group.

*Table 4.4: Mean Differences Between Churned and Retained Customers by Satisfaction Group*

| | Satisfaction group | | |
|---|---|---|---|
| Variable | Low | Medium | High |
| Tenure (months) | −6.65 | −7.47 | −8.66 |
| Cashback received | −17.24 | −20.98 | −22.67 |
| Days since last order | −1.18 | −1.45 | −2.07 |
| Warehouse-to-home distance | 0.42 | 1.68 | 2.29 |
| Registered addresses | 0.42 | 0.09 | 0.49 |
| Order count | 0.34 | −0.13 | −0.85 |
| Coupons redeemed | 0.37 | −0.01 | −0.37 |
| Time spent on app (hours) | 0.00 | 0.01 | 0.08 |

Three variables separate the outcome groups in the same direction across all three segments, and in each case the gap widens as satisfaction rises. Customer relationship length shows the largest difference, from 6.65 months in the low satisfaction group to 8.66 months in the high satisfaction group. Cashback received follows the same pattern, from 17.24 to 22.67. Days since last order is also consistently lower among churned customers.

WarehouseToHome is higher among churned customers in every group, and the difference is five times larger in the high satisfaction group than in the low satisfaction group. Order count and coupons redeemed change direction across the scale, from slightly higher among churned customers at low satisfaction to lower at high satisfaction. Time spent on the app shows almost no separation in any group.

Figure 4.5 shows how the recorded churn rate varies across the range of the four variables with the largest differences. Cashback received, WarehouseToHome and NumberOfAddress each vary approximately linearly with churn rate, while customer relationship length does not: customers in the lowest decile record a churn rate above 50%, compared with under 15% across all remaining deciles.

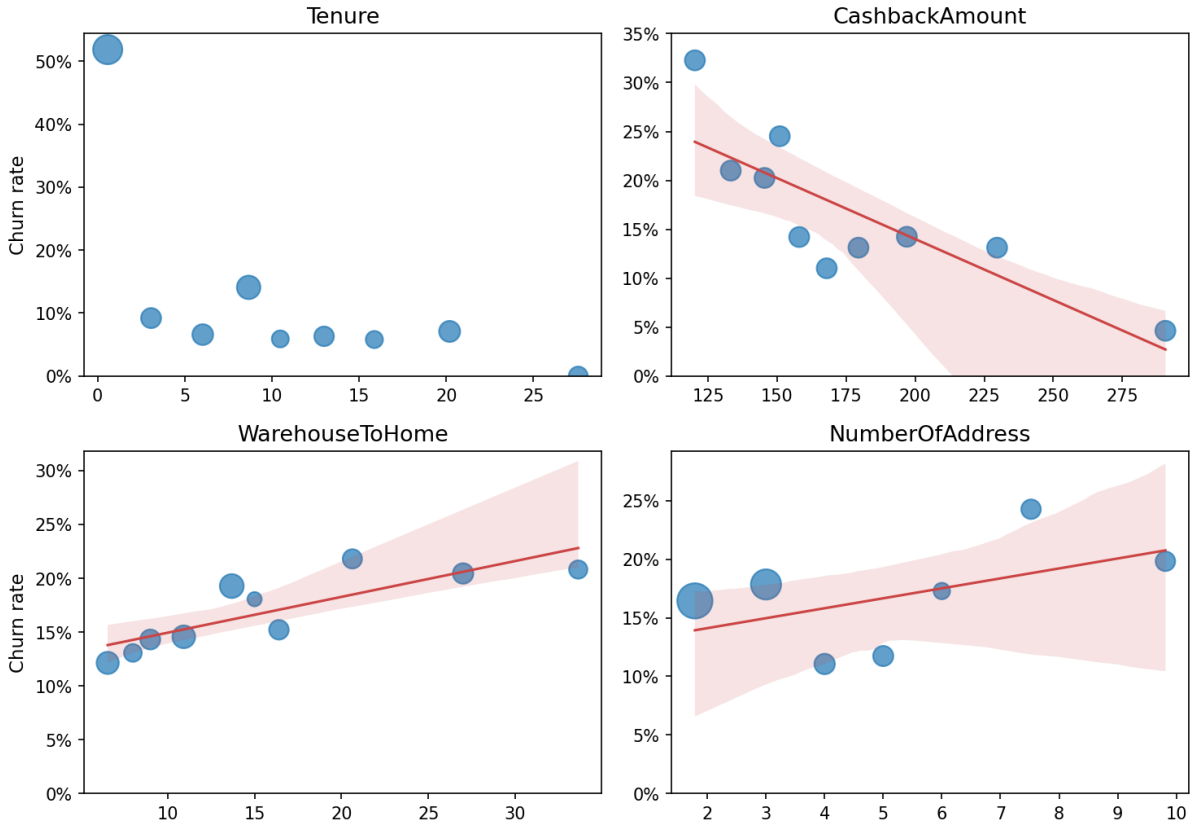



*Figure 4.5: Recorded Churn Rate Across the Range of Four Selected Variables*

Note. Each point represents one decile of the variable's distribution, plotted at the decile mean. Point size is proportional to the number of customers in that decile. Trend lines show a fitted linear relationship with a 95% confidence band. No line is shown for Tenure, where the relationship is not linear. Variable names correspond to those defined in Table 4.1.

### 4.3.4 Correlation Analysis

Figure 4.6 reports Spearman correlations between each variable and the recorded churn indicator, for the pooled sample and within each satisfaction group. Spearman was used because several variables are discrete or non-normally distributed. Coefficients are reported for the numeric and binary variables only, since the categorical predictors are not encoded in the descriptive analyses.




*Figure 4.6: Spearman Correlations with Recorded Churn Across the Pooled Sample and Satisfaction Groups*

Note. Variable names correspond to those defined in Table 4.1.

Four variables show consistent associations across all three groups. Tenure has the strongest relationship in every group and strengthens as satisfaction rises, from −0.304 in the low satisfaction group to −0.486 in the high satisfaction group. Complain is positively associated with churn throughout, most strongly in the medium satisfaction group at 0.305. DaySinceLastOrder and CashbackAmount are both negatively associated in every group, and both also strengthen with satisfaction.

The remaining coefficients are all below 0.15 in absolute value. NumberOfDeviceRegistered and CityTier remain weakly positive in all three groups. Others vary across the scale. WarehouseToHome rises from 0.033 in the low satisfaction group to 0.146 in the high satisfaction group, while OrderAmountHikeFromlastYear changes sign, from −0.071 to 0.115. OrderCount and CouponUsed also change sign, moving from slightly positive at low

satisfaction to slightly negative at high satisfaction. The associations among the predictors themselves are shown in Figure 4.7.

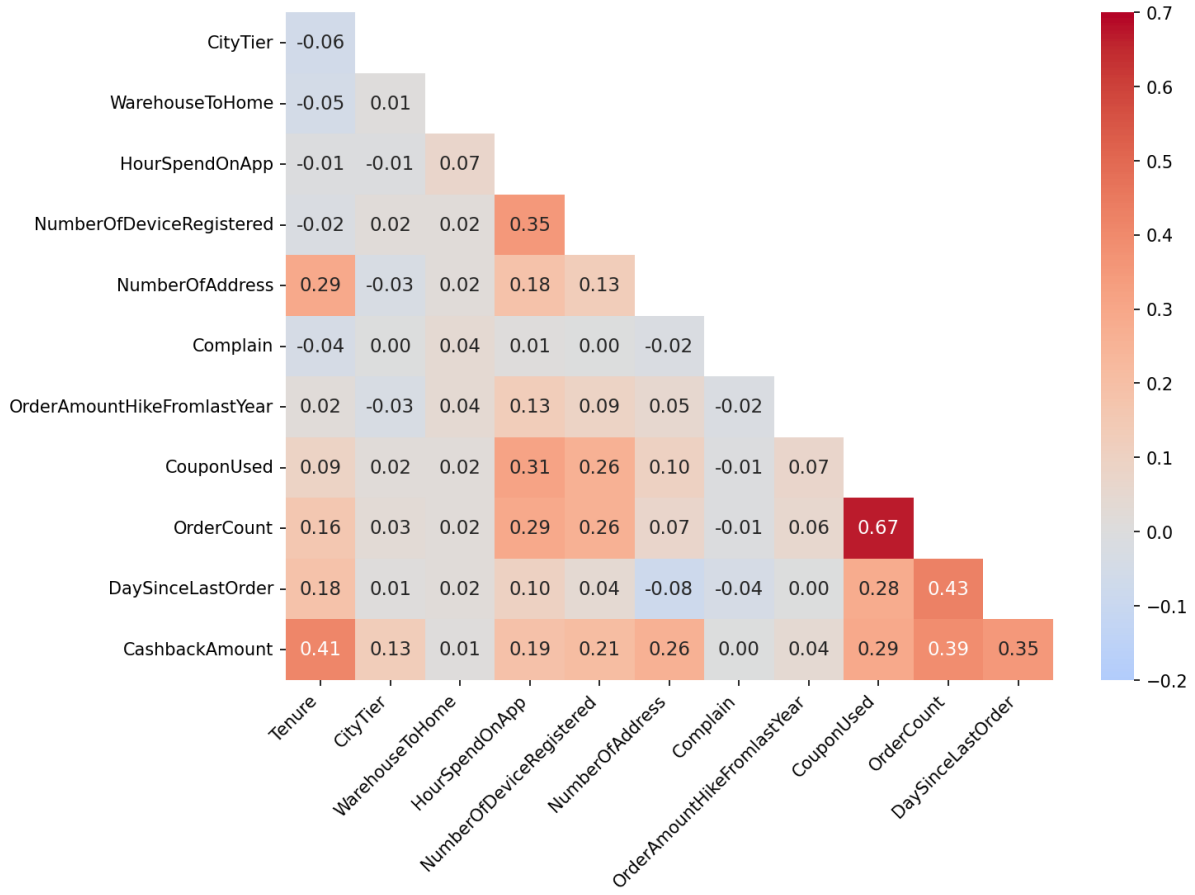



*Figure 4.7: Spearman Correlations Among the Predictors*

Note. Only the lower triangle is shown, as the matrix is symmetric. Variable names correspond to those defined in Table 4.1.

CouponUsed and OrderCount are strongly related at 0.67, while OrderCount is also related to DaySinceLastOrder at 0.43 and to CashbackAmount at 0.39. CashbackAmount and Tenure are related at 0.41. Complain, by contrast, is uncorrelated with every other variable. Several predictors therefore carry overlapping information, and the coefficients in Figure 4.6 cannot be read as independent contributions.

## 4.4 XGBoost-Based Churn Modelling Across Satisfaction Groups

### 4.4.1 Modelling Rationale and Specification

XGBoost classification models were estimated to examine whether the predictive signals associated with recorded churn differ across the satisfaction groups. A pooled model and three satisfaction-specific models were fitted using the same candidate predictors, with Churn as the outcome and CustomerID and SatisfactionScore excluded for the reasons given in Sections 3.3.2 and 3.4.1. No feature selection was performed. Selecting variables separately within each group according to their observed association with churn would have predetermined which features were available to each model, making later differences in SHAP rankings partly attributable to different model inputs rather than to differences between the groups. One-hot encoding produced 24 features from the 17 candidate predictors.

Preprocessing was implemented within a scikit-learn Pipeline using a ColumnTransformer, with imputation and encoding fitted separately on each cross-validation training fold as described in Section 3.4.1. The models used 200 estimators, a maximum tree depth of 4 and a learning rate of 0.1, with the binary:logistic objective and logloss as the evaluation metric. For each analytical sample, scale_pos_weight was calculated from the within-sample ratio of retained to churned customers and held constant across its five folds. A fixed random seed of 42 was used. No hyperparameter tuning was carried out, and the same specification was applied to all four models to preserve comparability across analytical samples.

### 4.4.2 Cross-Validated Model Performance

*Table 4.5: Cross-Validated Performance of the Pooled and Satisfaction-Specific XGBoost Models*

| | | Churn | | ROC-AUC | | AP | | F1 | |
|---|---|---|---|---|---|---|---|---|---|
| Analytical sample | *n* | *n* | % | Mean | SD | Mean | SD | Mean | SD |
| Pooled | 5,630 | 948 | 16.8 | 0.9709 | 0.0060 | 0.8895 | 0.0174 | 0.7991 | 0.0256 |
| Low | 1,750 | 208 | 11.9 | 0.9834 | 0.0108 | 0.9417 | 0.0243 | 0.8909 | 0.0163 |
| Medium | 2,772 | 476 | 17.2 | 0.9849 | 0.0058 | 0.9357 | 0.0103 | 0.8860 | 0.0252 |
| High | 1,108 | 264 | 23.8 | 0.9942 | 0.0055 | 0.9849 | 0.0137 | 0.9661 | 0.0150 |

Note. Values are means and standard deviations across five stratified validation folds. The no-skill baseline for Average Precision corresponds to the churn prevalence shown in the Churn % column. F1 was calculated using a classification threshold of 0.5.

Table 4.5 reports the mean and standard deviation across five stratified validation folds, following the evaluation procedure described in Section 3.4.3. All four models separated churned from retained customers with ROC-AUC above 0.97. The satisfaction-specific values ranged from 0.9834 in the low satisfaction group to 0.9942 in the high satisfaction group, with fold-level standard deviations of 0.011 or below, indicating that discrimination was consistent across folds rather than produced by a single favourable partition. The low satisfaction model showed the greatest fold-level variability, which is consistent with its smaller number of churn cases.

Average Precision exceeded the corresponding no-skill baseline in every sample, ranging from 0.8895 to 0.9849 against churn prevalence baselines of 0.119 to 0.238. F1 ranged from 0.886 to 0.966 across the segment models. The pooled model recorded the lowest AP and F1, at 0.8895 and 0.7991, while the high satisfaction model recorded the highest values on all three metrics despite being fitted to the smallest sample. Because the four models were evaluated on different analytical samples, these differences are descriptive and no test of between-model performance was conducted; segment-specific estimation was adopted to allow predictor importance to vary across groups rather than to improve predictive accuracy.

The resulting values are unusually high for a customer-level churn task. Section 4.4.3 therefore examines whether preprocessing, duplication or the analytical procedure could account for them before the models are interpreted using SHAP in Section 4.5.

### 4.4.3 Robustness Checks and Performance Diagnostics

The ROC-AUC values reported above are high enough to require confirmation that they arise from the data rather than from the analytical procedure. Following the cross-validated evaluation, five diagnostics were conducted, each addressing a distinct mechanism of artificial inflation. Summaries are reported in Table 4.6.

The first tested whether the pipeline could produce above-chance performance once the predictor–outcome relationship was removed. The Churn label was randomly permuted ten times within each satisfaction segment and the models re-evaluated using the same stratified five-fold procedure. Mean ROC-AUC fell to chance level in every segment, confirming that the pipeline retains no predictive performance when the original association is broken.

The second compared preprocessing configurations. Fitting imputation and encoding to the complete sample before cross-validation, using identical fold indices, changed ROC-AUC by no more than 0.005 in any model. The reported performance is therefore not attributable to preprocessing fitted outside the validation loop.

The third examined whether performance depended on a small number of leading predictors. Three ablation configurations were evaluated, excluding Tenure alone, Complain alone, and both jointly. Excluding both left ROC-AUC above 0.93 in every segment, while F1 declined more noticeably, indicating that the two features contributed materially to classification at the 0.5 threshold without accounting for the broader ranking performance.

The fourth assessed whether performance saturated at small training sizes, which would suggest a near-deterministic labelling rule. A learning curve was estimated for the pooled model at five training sizes, from 225 to 4,504 observations, comparing training and validation ROC-AUC. Mean validation ROC-AUC rose steadily with training size while training ROC-AUC remained at or near 1.000, narrowing the gap (Figure 4.8).

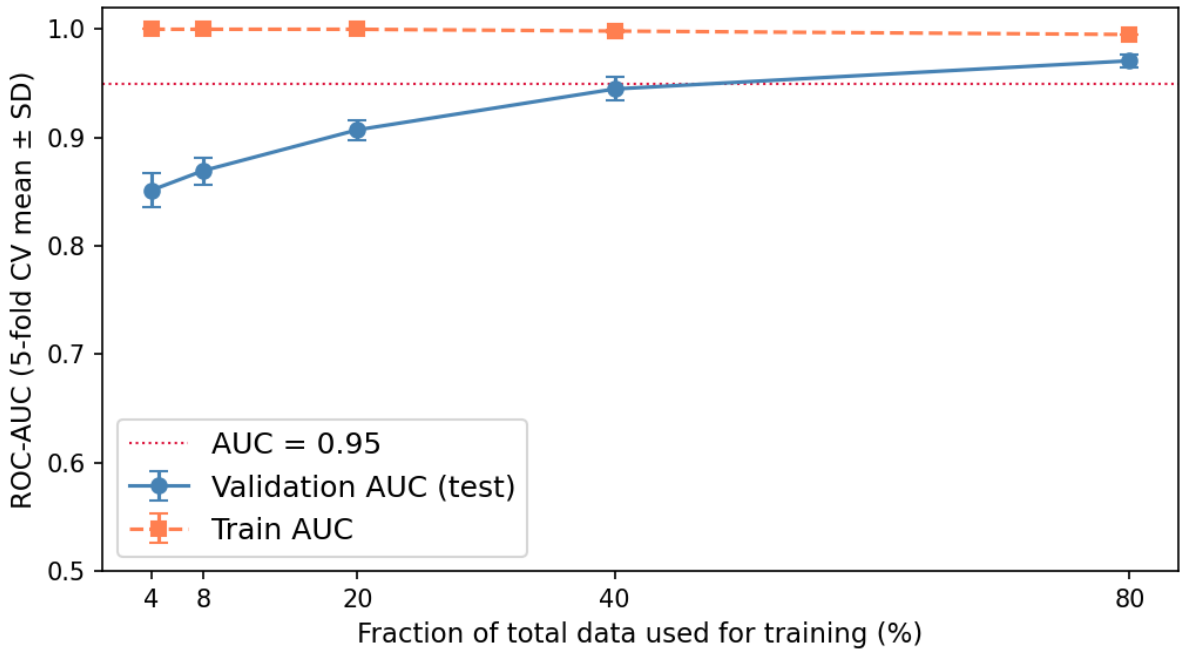



*Figure 4.8: Learning Curve for the Pooled XGBoost Model*

Note. Points are means across five stratified folds and error bars show one standard deviation. Training sizes are expressed as a percentage of the full sample; the maximum of 80% is the largest training set available under five-fold cross-validation.

The fifth examined duplicate records. After excluding CustomerID, 557 rows shared duplicate analytical profiles, and no duplicate group contained conflicting Churn labels. Retaining one copy of each profile left ranking performance essentially unchanged but reduced F1 in the medium and high satisfaction groups. Duplicates were retained in the main analysis because the dataset does not establish that rows with identical recorded attributes represent the same customer.

*Table 4.6: Summary of Robustness Diagnostics and the Unverifiable Temporal Ordering of the Outcome*

| Diagnostic | Risk examined | Key evidence | Interpretation |
|---|---|---|---|
| 1. Shuffled target | Embedded target signal | Mean ROC-AUC fell to 0.497, 0.498 and 0.509 in the low, medium and high satisfaction groups | Excluded |
| 2. Preprocessing comparison | Fold leakage | Difference in ROC-AUC below 0.005 across all models; 0.0002 in the pooled model | Excluded |
| 3. Feature exclusion | Dominance of Tenure and Complain | ROC-AUC remained at 0.9769, 0.9394 and 0.9862 after excluding both; medium F1 fell from 0.886 to 0.724 | Excluded |
| 4. Learning curve | Immediate small-sample saturation | Validation ROC-AUC rose from 0.851 (n = 225) to 0.971 (n = 4,504); the training–validation gap narrowed from 0.149 to 0.024 | Excluded |
| 5. Duplicate analysis | Memorisation through duplicate rows | 557 duplicate rows (9.89%), all with consistent Churn labels; ROC-AUC changed by no more than 0.004, though F1 fell by 0.022 (medium) and 0.044 (high) | Limited effect on ranking; some effect on threshold classification |
| Temporal ordering | Outcome-proximal measurement | No churn date, observation window or predictor timestamps available | Not excluded; cannot be verified |

None of the five diagnostics identified a mechanical source of artificial inflation. The results support the presence of strong predictive structure within the dataset, but do not establish that comparable performance would be achieved in a deployment setting. Therefore, the remaining consideration concerns what the outcome records rather than how the models were fitted. The implications are discussed in Section 5.2.2.

### 4.4.4 Preliminary Feature Importance from XGBoost

Figure 4.9 reports the ten highest gain-based importance scores within each model. Tenure ranked first in all four, with normalised scores of 0.151 in the pooled model, 0.166 in the low satisfaction model, 0.133 in the medium satisfaction model and 0.176 in the high satisfaction model. Complain and the Laptop & Accessory order category indicator appeared among the ten leading predictors in every model.

The ranking below Tenure differed across the models. Coupons used entered the leading ten in the medium satisfaction model but not in the low or high satisfaction models, and number of addresses ranked fifth in the high satisfaction model but did not appear in the medium satisfaction ranking. These differences provide preliminary evidence that the fitted models relied on different combinations of predictors across the satisfaction groups.

Gain-based importance is reported here as an initial diagnostic only. It measures each feature's contribution to loss-reducing splits as a single unsigned magnitude and does not indicate whether higher feature values push predictions towards churn or retention. Section 4.5 therefore uses SHAP as the main interpretive basis, since it assigns each feature a signed contribution and uniquely satisfies consistency within the class of additive feature attribution methods (Lundberg and Lee, 2017; Lundberg et al., 2020).

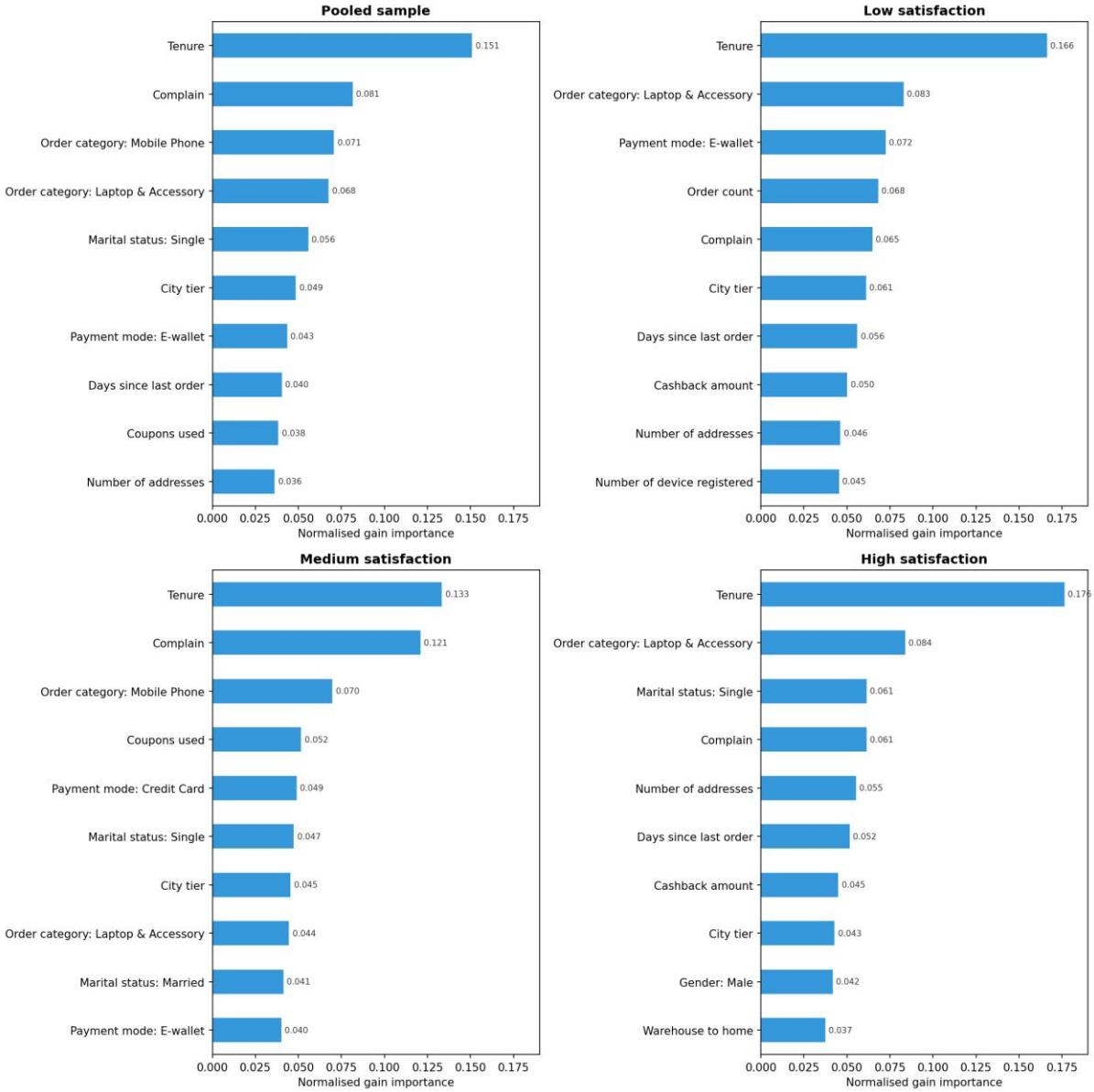



*Figure 4.9: Gain-Based XGBoost Feature Importance Across Satisfaction Groups*

Note. Bars show the ten highest normalised gain importance scores within each model. Scores are interpreted within each model and do not indicate whether higher feature values increase or decrease predicted churn.

## 4.5 SHAP-Based Interpretation of Churn Drivers Across Satisfaction Groups

### 4.5.1 Purpose and Interpretation of SHAP Analysis

Following the cross-validated assessment reported in Section 4.4.2, final XGBoost pipelines were fitted to the complete pooled sample and separately to the complete sample within each satisfaction segment, and SHAP values were computed for the same observations used to fit each model, as described in Section 3.4.4. The resulting values are in-sample explanations of the fitted models rather than out-of-fold explanations based on held-out observations.

For a given observation, a positive SHAP value indicates that the observed feature value increases the model output relative to its baseline and pushes the prediction towards the churn class, while a negative value pushes it towards the retained class. Overall importance is summarised using mean absolute SHAP values. These values describe how each fitted model used the predictors and are not interpreted as causal effects.

### 4.5.2 SHAP Feature Importance and Direction Within Satisfaction Segments

Figures 4.10 to 4.12 present the SHAP summary plots for the low, medium and high satisfaction models. Features are ordered by mean absolute SHAP value, and the horizontal distribution of observations shows whether particular feature values pushed predictions towards churn or retention. The pooled model is reported in Table 4.7 as an aggregate reference.

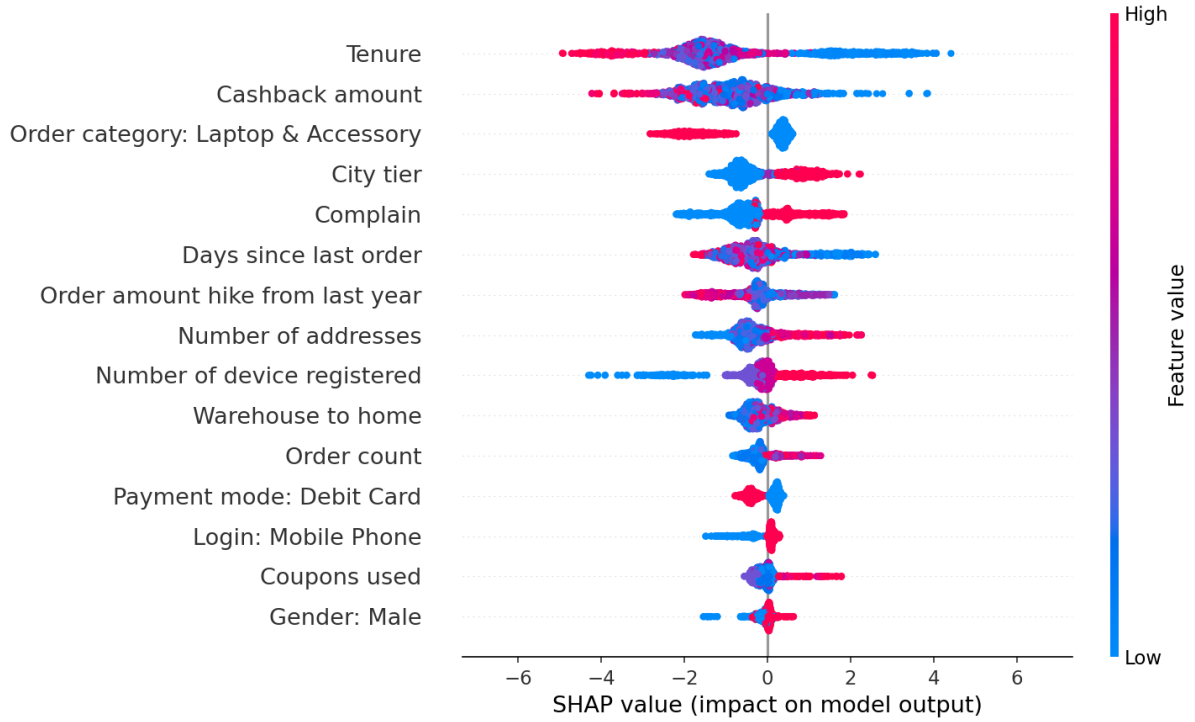



*Figure 4.10: SHAP Summary Plot for the Low-Satisfaction Model*

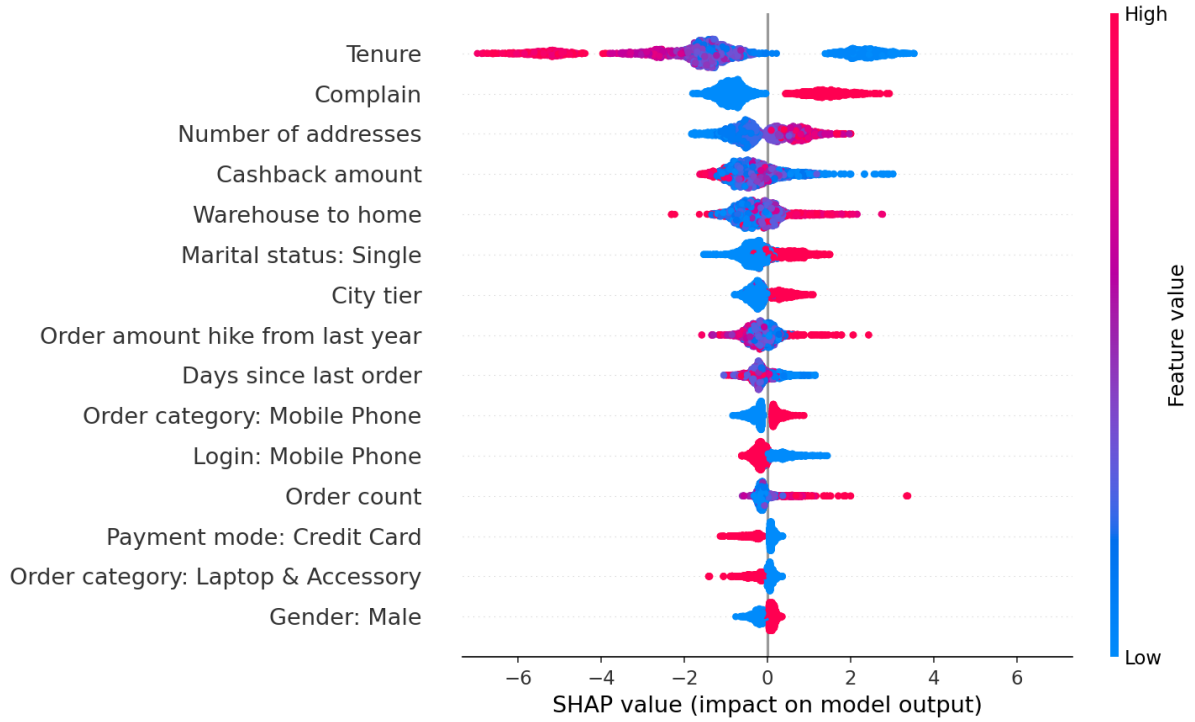



*Figure 4.11: SHAP Summary Plot for the Medium-Satisfaction Model*

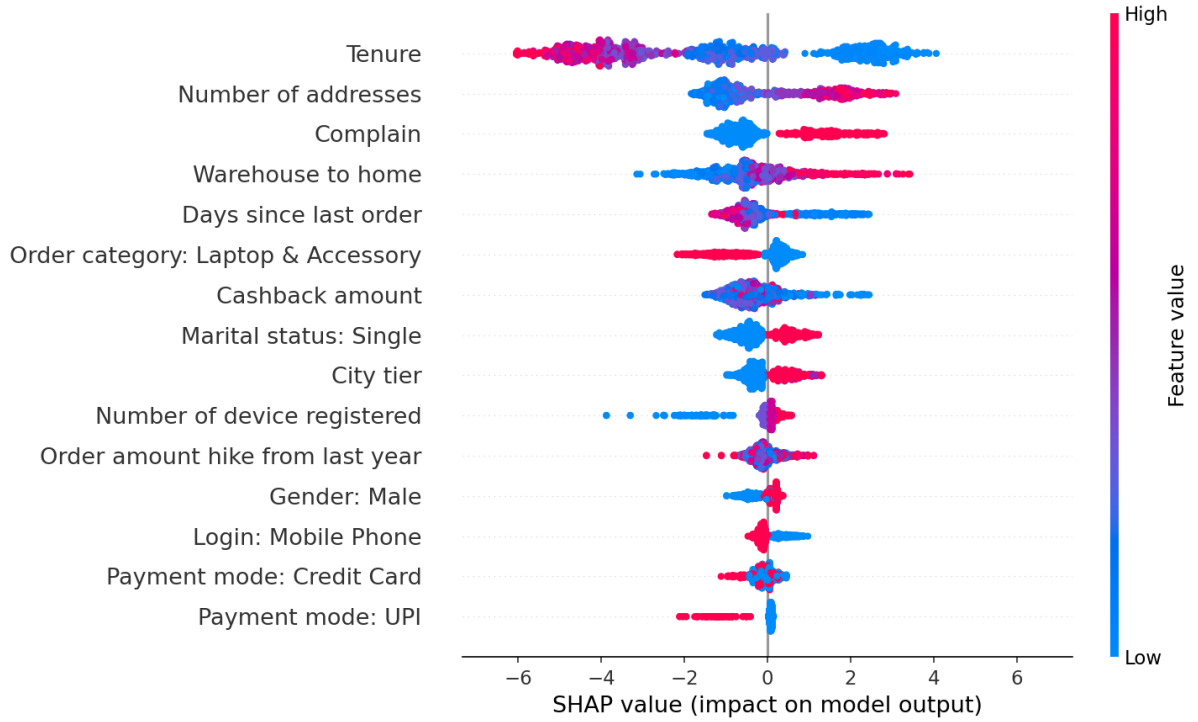



*Figure 4.12: SHAP Summary Plot for the High-Satisfaction Model*

Note. Red points represent higher feature values and blue points lower values. Positive SHAP values push predictions towards churn, negative values towards retention. The horizontal axis range is common to all three plots. Variable names correspond to those defined in Table 4.1.

Tenure ranked first in all three segment models, with mean absolute SHAP values of 1.81, 2.21 and 2.74 in the low, medium and high satisfaction models. Lower values produced positive contributions and pushed predictions towards churn, while higher values pushed them towards retention.

Complain showed a consistent direction across the three models, with a recorded complaint producing positive contributions. Its relative position varied: it ranked fifth in the low satisfaction model, behind CashbackAmount, the Laptop & Accessory category indicator and CityTier, but second in the medium satisfaction model and third in the high satisfaction model.

CashbackAmount also retained a consistent direction, with lower values pushing predictions towards churn. Its relative importance declined as satisfaction increased, from second in the low satisfaction model to fourth in the medium and seventh in the high satisfaction model.

CityTier ranked fourth in the low satisfaction model, with a mean absolute SHAP value of 0.706 against 0.240 in the pooled model, and higher city tiers pushing predictions towards churn. Its position was less stable than the leading features, as reported in Section 4.5.3.

The high satisfaction model showed the most distinctive hierarchy. NumberOfAddress ranked second, with higher values pushing predictions towards churn, and WarehouseToHome ranked fourth, also in the churn direction. Both occupied lower positions in the low satisfaction model, at eighth and tenth, but ranked third and fifth in the medium satisfaction model.

Table 4.7 reports the feature rankings and predominant directions across the four models. Among the leading features, the differences between segments lay in the relative ordering of predictors rather than in reversals of direction. The only directional inconsistency occurred in OrderAmountHikeFromlastYear, which ranked seventh to thirteenth across the models and reversed sign between the low satisfaction segment and the other two, consistent with the correlation pattern reported in Section 4.3.4.

*Table 4.7: SHAP Feature Rankings and Predominant Directions Across the Pooled and Satisfaction-Specific Models*

| Feature | Direction | Pooled | Low | Medium | High |
|---|---|---|---|---|---|
| Tenure | Lower → churn | 1 | 1 | 1 | 1 |
| Complain | Present → churn | 2 | 5 | 2 | 3 |
| NumberOfAddress | Higher → churn | 3 | 8 | 3 | 2 |
| CashbackAmount | Lower → churn | 4 | 2 | 4 | 7 |
| DaySinceLastOrder | Lower → churn | 6 | 6 | 9 | 5 |
| Laptop & Accessory | Present → retention | 5 | 3 | 14 | 6 |
| WarehouseToHome | Higher → churn | 7 | 10 | 5 | 4 |
| MaritalStatus: Single | Present → churn | 8 | 18 | 6 | 8 |
| CityTier | Higher → churn | 9 | 4 | 7 | 9 |
| NumberOfDeviceRegistered | Higher → churn | 10 | 9 | 16 | 10 |
| OrderAmountHikeFromlastYear | Varies by segment | 13 | 7 | 8 | 11 |
| PreferedOrderCat: Mobile Phone | Present → churn | 19 | 20 | 10 | 19 |

Note. Rank 1 indicates the highest mean absolute SHAP value within a model. The table includes features ranked among the ten most influential in at least one model, so a feature may rank considerably lower in the remaining models. Rankings are relative within each model and are not directly comparable in absolute magnitude across models. Direction refers to the predominant SHAP direction observed in the summary plots.

### 4.5.3 Bootstrap Stability of SHAP Feature Rankings

Table 4.8 reports the mean rank and rank standard deviation of the eight highest-ranked features within each satisfaction segment across 50 bootstrap resamples, following the procedure described in Section 3.4.4. Lower standard deviations indicate that a feature's position was less sensitive to the resampled observations.

The leading features were stable in every segment. Tenure held a mean rank of 1.00 with zero variation throughout. The segment-specific patterns reported in Section 4.5.2 were also supported: CashbackAmount retained a mean rank of 2.02 in the low satisfaction segment, Complain remained second with zero variation in the medium satisfaction segment, and NumberOfAddress remained second in the high satisfaction segment at 2.04.

WarehouseToHome held fourth position in the high satisfaction model at 4.16. All of these were classified as stable.

Variability was concentrated among the mid-ranked features. In the low satisfaction segment, CityTier moved from fourth in the fitted model to a mean rank of 6.72, with a standard deviation of 1.980. The Laptop & Accessory indicator and Complain recorded closely matched mean ranks of 3.94 and 4.20, with standard deviations of 1.490 and 1.212. The ordering among these three features should therefore not be interpreted as fixed.

The medium satisfaction segment showed the sharpest division: its five leading features all recorded standard deviations below 0.60, while every remaining feature exceeded 1.5. Stability within that leading group does not, however, fix the order inside it. NumberOfAddress and CashbackAmount held mean ranks of 3.46 and 3.54, a difference of 0.08, so their relative position is not established.

The principal cross-segment patterns reported in Section 4.5.2 are therefore robust to sampling variation, while differences between closely ranked or lower-ranked predictors are not.

*Table 4.8: Bootstrap Stability of SHAP Feature Rankings (50 Resamples, Eight Highest-Ranked Features per Segment)*

| Feature | Mean rank | SD of rank | Stability |
|---|---|---|---|
| ***Low satisfaction*** | | | |
| Tenure | 1.00 | 0.000 | Stable |
| CashbackAmount | 2.02 | 0.141 | Stable |
| PreferedOrderCat: Laptop & Accessory | 3.94 | 1.490 | Moderate |
| Complain | 4.20 | 1.212 | Moderate |
| DaySinceLastOrder | 5.44 | 1.327 | Moderate |
| CityTier | 6.72 | 1.980 | Unstable |
| NumberOfAddress | 6.82 | 1.380 | Moderate |
| OrderAmountHikeFromlastYear | 7.64 | 1.914 | Unstable |
| ***Medium satisfaction*** | | | |
| Tenure | 1.00 | 0.000 | Stable |
| Complain | 2.00 | 0.000 | Stable |
| NumberOfAddress | 3.46 | 0.503 | Stable |
| CashbackAmount | 3.54 | 0.503 | Stable |
| WarehouseToHome | 5.40 | 0.571 | Stable |
| MaritalStatus: Single | 6.88 | 1.649 | Unstable |
| DaySinceLastOrder | 7.76 | 1.572 | Unstable |
| OrderAmountHikeFromlastYear | 8.46 | 1.764 | Unstable |
| ***High satisfaction*** | | | |
| Tenure | 1.00 | 0.000 | Stable |
| NumberOfAddress | 2.04 | 0.198 | Stable |
| Complain | 3.22 | 0.648 | Stable |
| WarehouseToHome | 4.16 | 0.842 | Stable |
| DaySinceLastOrder | 5.50 | 1.055 | Moderate |
| PreferedOrderCat: Laptop & Accessory | 6.52 | 1.460 | Moderate |
| CashbackAmount | 6.68 | 1.435 | Moderate |
| MaritalStatus: Single | 7.54 | 1.129 | Moderate |

## 4.6 RFM-Informed Behavioural Profiling

### 4.6.1 Profile Construction

The framework described in Section 3.4.5 was applied using three proxy variables: days since last order for recency, number of orders for frequency, and cashback received for monetary value. The dataset records no expenditure or transaction revenue, so cashback is used as the available indicator of monetary value; its limitations are set out in Section 3.6. Table 4.9 reports the resulting bands.

Recency and monetary value were divided at the terciles of the pooled distribution. Neither variable has a documented time base or spending interpretation that would support a substantively defined cutpoint, so an equal-frequency split was preferred to an imposed threshold. Number of orders was banded by rule at one, two, and three or more, because it is a discrete count with heavy ties: 1,751 customers placed one order and 2,283 placed two, so quantile cutting would assign customers with identical order counts to different bands. The rule still produces bands of comparable size.

Three bands were used on each dimension rather than five. The dimensions combine multiplicatively, so finer banding would leave too few customers per profile for the segment-level comparison reported in Section 4.6.3. Profiles containing fewer than 30 customers were excluded from that comparison.

Thresholds were derived from the pooled sample and applied unchanged within each satisfaction segment, so that a code represents the same behavioural range in all three groups. Recency was reverse-coded, so that R3 identifies the most recent customers. The codes record behavioural position rather than risk: a higher R, F or M value indicates more recent activity, more orders or higher monetary value, and carries no assumption about churn.

*Table 4.9: Definition and Operational Ranges of the RFM Profile Bands*

| Dimension | Proxy variable | Band | Operational range | n |
|---|---|---|---|---|
| | | R1 least recent | 6–46 days | 1,862 |
| Recency (R) | DaySinceLastOrder | R2 | 3–5 days | 1,866 |
| | | R3 most recent | 0–2 days | 1,902 |
| | | F1 low | 1 order | 1,751 |
| Frequency (F) | OrderCount | F2 moderate | 2 orders | 2,283 |
| | | F3 high | 3 or more orders | 1,596 |
| | | M1 low | 0.00–150.00 | 1,880 |
| Monetary (M) | CashbackAmount | M2 moderate | 150.01–181.05 | 1,874 |
| | | M3 high | 181.06–324.99 | 1,876 |

## 4.6.2 RFM Dimension-Level Churn Patterns

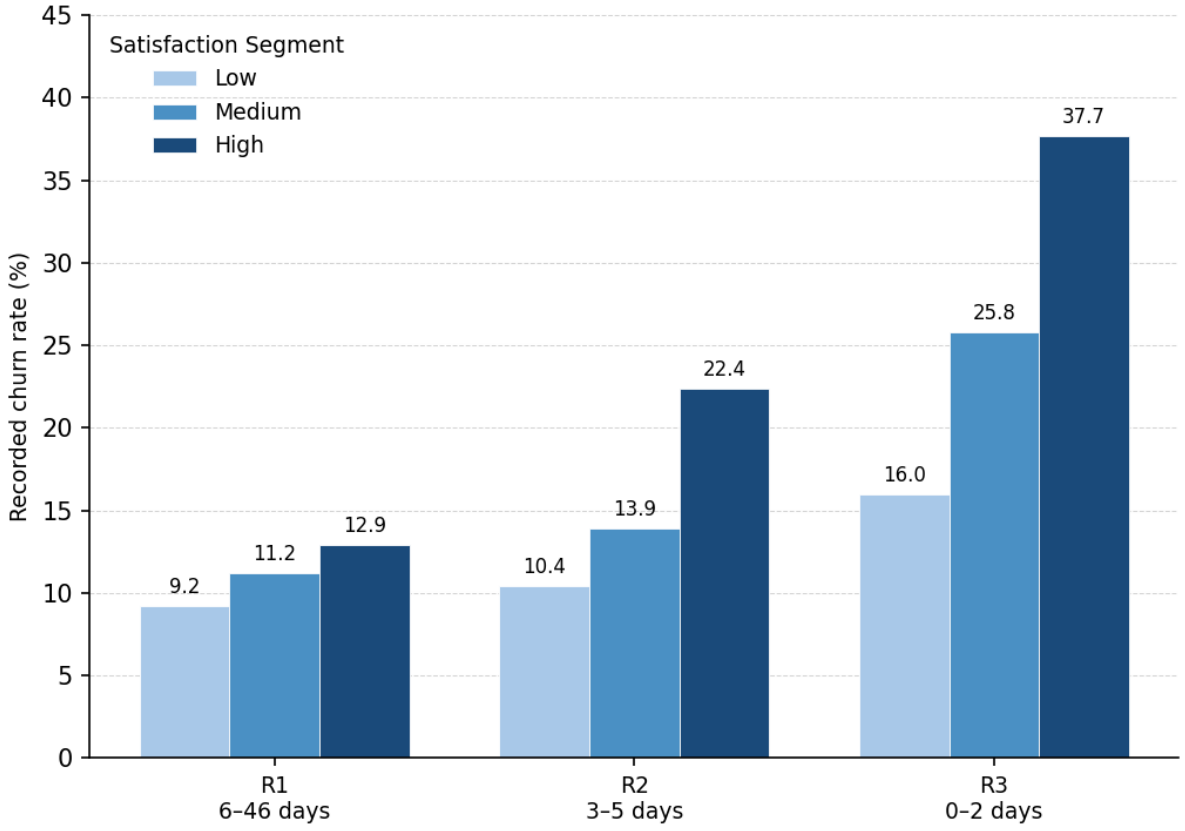



*Figure 4.13: Recorded Churn Rate by Recency Band and Satisfaction Segment*

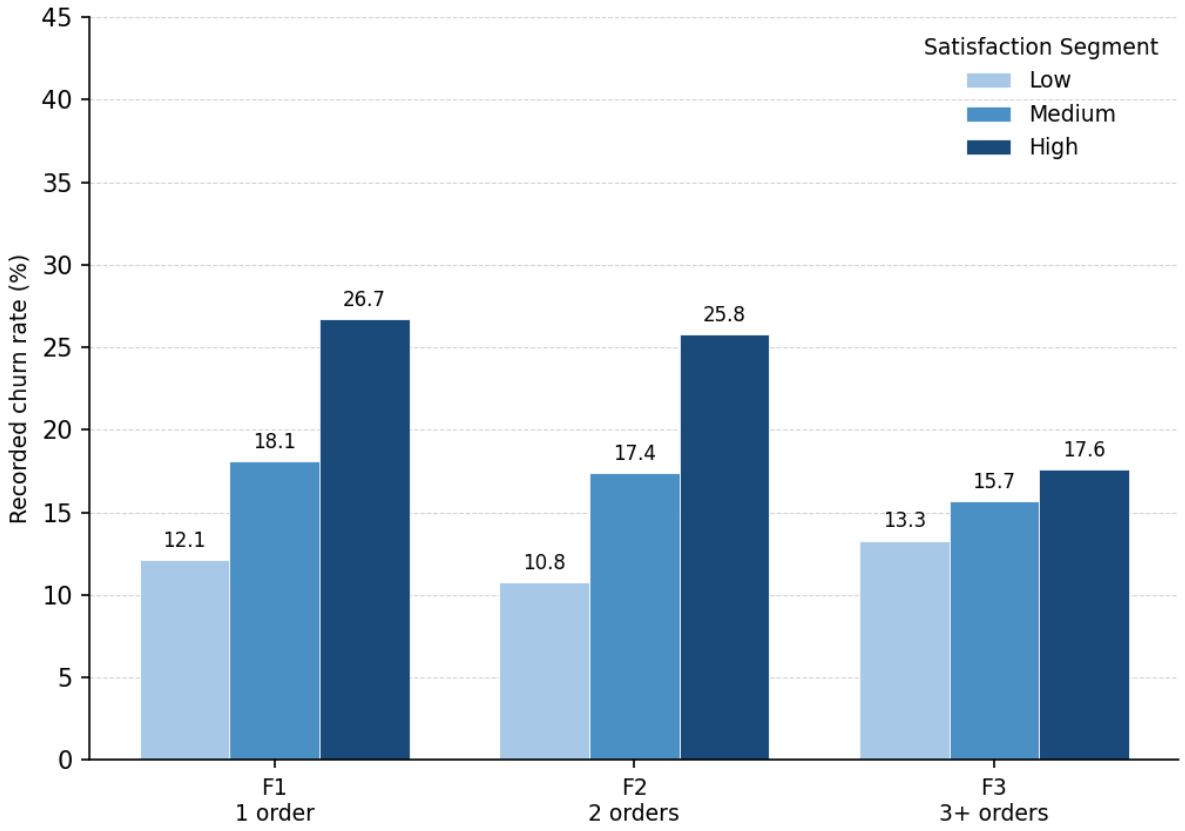



*Figure 4.14: Recorded Churn Rate by Frequency Band and Satisfaction Segment*

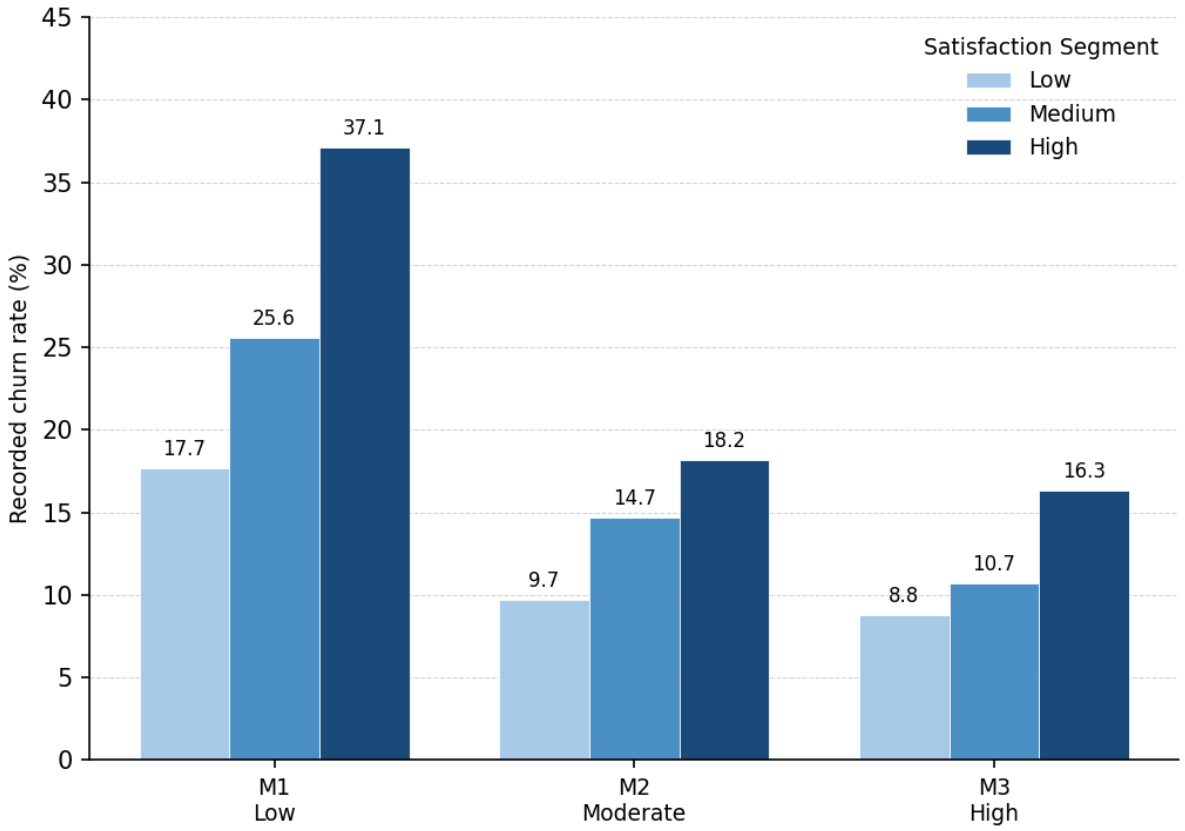



*Figure 4.15: Recorded Churn Rate by Monetary Band and Satisfaction Segment*

Figures 4.13 to 4.15 report the recorded churn rate within each band, calculated separately for the three satisfaction segments.

Monetary value separated the outcome groups most clearly (Figure 4.15). Churn declined from the lowest to the highest band in every segment, from 17.7% to 8.8% at low satisfaction, 25.6% to 10.7% at medium and 37.1% to 16.3% at high.

Recency ran counter to conventional RFM logic (Figure 4.13). The most recent band recorded the highest churn rate in every segment, at 16.0%, 25.8% and 37.7%, against 9.2%, 11.2% and 12.9% in the least recent band. This follows from the operationalisation set out in Section 3.3.1. Because the outcome records a status rather than an inactivity-derived event, recent transacting and a recorded churn status are not mutually exclusive, and the recency dimension does not carry its conventional meaning in this dataset.

Frequency showed the weakest pattern (Figure 4.14). The one-order and two-order bands recorded almost identical rates within each segment, differing by 1.3 percentage points or less. The direction was also inconsistent: the highest-frequency band recorded the lowest rate at high satisfaction, at 17.6% against 26.7% and 25.8%, but the highest rate at low satisfaction, at 13.3% against 12.1% and 10.8%.

### 4.6.3 Cross-Segment Comparison of Combined RFM Profiles

The three dimensions were combined into profile codes. Sixty segment-profile cells contained at least 30 customers and were retained for comparison, covering 24 profiles. Figure 4.16 reports the recorded churn rate for each retained cell.

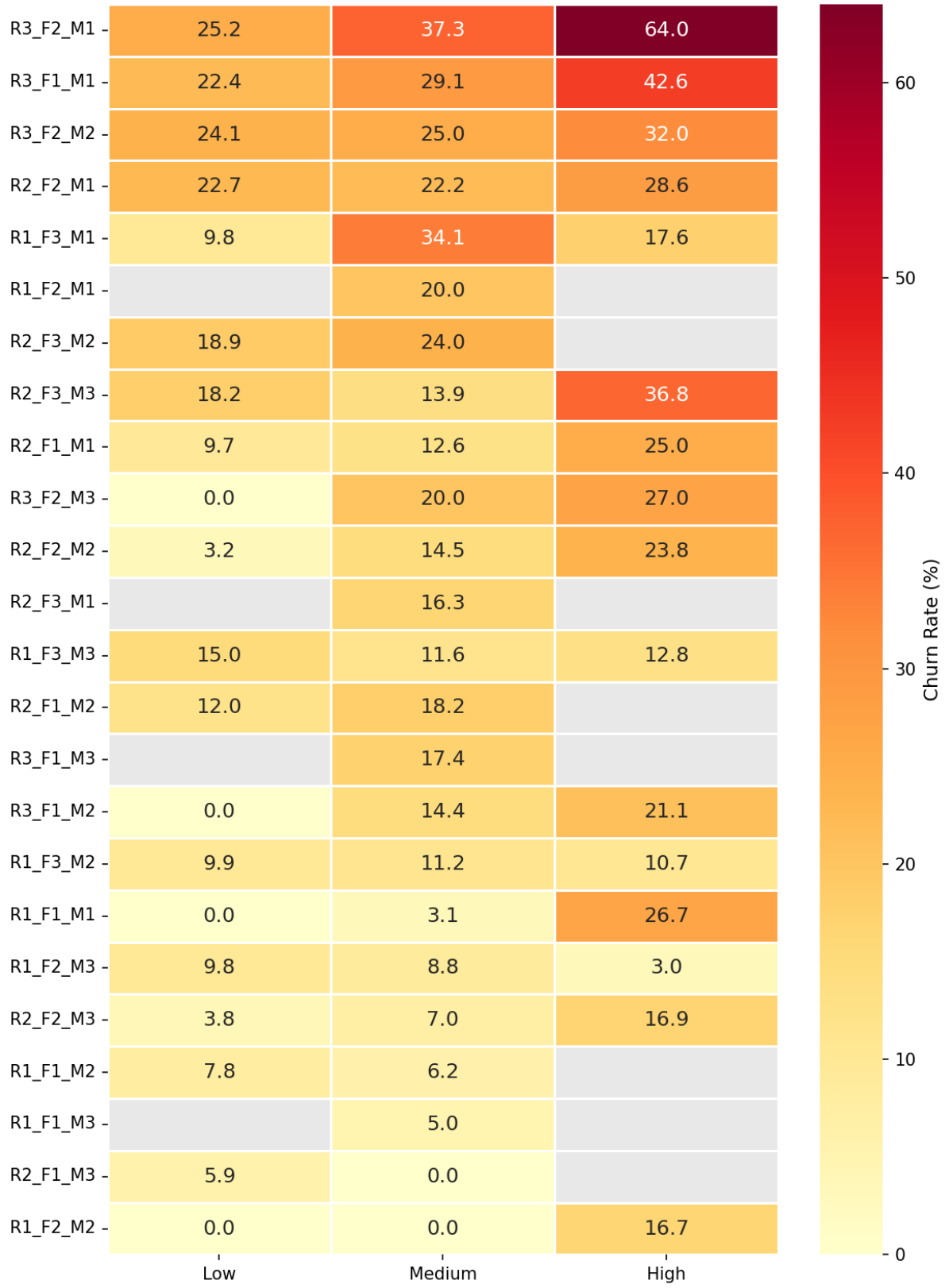



*Figure 4.16: Recorded Churn Rate by Combined RFM Profile and Satisfaction Segment*

Note. Grey cells indicate that the profile contained fewer than 30 customers in that satisfaction segment and was excluded from comparison. A value of 0.0 indicates that no customer in that cell was recorded as churned. Profiles are ordered by their churn rate in the pooled sample.

The four profiles with the highest pooled churn rates all combined one or two orders with monetary value in the lower two bands, and none involved the highest monetary band. The dimension-level results in Section 4.6.2 showed monetary value to be the clearest single separator and frequency the weakest, but the leading combined profiles paired both, indicating that the two dimensions identify higher-churn customers jointly rather than through monetary value alone.

*Table 4.10: Recorded Churn Rate of R3_F2_M1 Relative to Satisfaction-Segment Baselines*

| Segment | Profile n | Churned | Churn rate | Wilson 95% CI | Segment baseline churn rate | Ratio to baseline |
|---|---|---|---|---|---|---|
| Low | 111 | 28 | 25.2% | 18.1 – 34.0% | 11.9% | 2.12 |
| Medium | 177 | 66 | 37.3% | 30.5 – 44.6% | 17.2% | 2.17 |
| High | 50 | 32 | 64.0% | 50.1 – 75.9% | 23.8% | 2.69 |

R3_F2_M1 recorded the highest rate in every segment. It identifies customers who last ordered within two days, placed two orders and received cashback in the lowest band. Its rate reached 64.0% among high satisfaction customers, against a segment baseline of 23.8%. Recent purchasing combined with limited frequency and low monetary value therefore accompanied a majority churn rate within the most satisfied group.

Table 4.10 also expresses the rate as a multiple of the corresponding segment baseline, since the three baselines differ. The multiple was similar in the low and medium segments, at 2.12 and 2.17, and higher in the high segment, at 2.69. The elevation associated with this profile was therefore not only a reflection of the higher baseline among satisfied customers.

The high satisfaction estimate rests on only 50 customers, although its Wilson interval does not overlap that of the medium satisfaction segment. R3_F2_M1 was also selected after

inspecting the observed results and no interaction test was conducted, so the comparison is exploratory. The adjacent profile R3_F1_M1, differing only in order count, stood at 1.88, 1.69 and 1.79 times its respective baselines and showed no comparable increase at high satisfaction, so the disproportionate elevation should not be generalised to low-monetary profiles as a class. The implications for retention priority are considered in Chapter 5.

# Chapter 5: Discussion and Managerial Implications

## 5.1 Chapter Introduction

This chapter interprets the findings reported in Chapter 4 and considers their implications for the research questions and CRM retention practice. Section 5.2 integrates the descriptive, predictive, SHAP and RFM-informed results to distinguish shared churn-related patterns from differences in predictor prominence across satisfaction segments. It also evaluates the unusually strong model performance and defines the boundaries within which the findings can be interpreted.

Section 5.3 then translates these findings into managerial implications. It first explains how prediction, SHAP, RFM profiles and satisfaction segmentation can support different stages of CRM decision-making. The discussion subsequently identifies shared retention priorities and develops differentiated responses for the low, medium and high satisfaction segments.

## 5.2 Discussion of the Main Findings

### 5.2.1 Descriptive Patterns Within Satisfaction Segments

To address RQ1 and RQ3 together, the descriptive and SHAP findings were considered in combination. The descriptive analysis showed how customers recorded as churned and retained differed on individual variables within each satisfaction segment. SHAP then indicated which predictors remained relatively influential when the full predictor set was considered jointly.

At the descriptive level, several patterns were shared across segments. Churned customers consistently had shorter customer tenure, lower cashback amount and lower days since last order, while delivery distance was higher in every segment, and registered addresses followed the same direction. Other variables were more segment-dependent. order count and coupons used were slightly higher among low-satisfaction churners but lower among high-satisfaction churners. App usage time showed almost no separation. The results therefore revealed some of the common descriptive core alongside variation in selected transactional and operational characteristics.

SHAP analysis of the fitted XGBoost models refined this interpretation. Customer tenure remained the most influential feature in every segment-specific model; below it, relative prominence differed by segment, as Table 4.7 sets out.

These findings do not imply that the three groups had entirely different churn factors. The main difference concerned how predictive information was distributed across a broadly shared set of characteristics. The descriptive differences observed for order count and coupons used, for example, did not translate into leading SHAP positions.

Variables that appear important when examined individually may behave differently once nonlinear relationships and interactions with other features are considered (Matuszelański and Kopczewska, 2022). SHAP distributes predictive attribution within the fitted multivariate model. A visible descriptive difference may therefore provide limited additional information when stronger or overlapping predictors are included (Lundberg et al., 2020). Order count and coupons used are themselves strongly correlated, and both overlap with predictors that ranked higher, so part of their descriptive difference is not independent information.

The bootstrap results provided further support for the main patterns while limiting overly precise interpretations. They reinforced the leading positions reported in Table 4.8, while variability concentrated among the mid-ranked features: city tier, in particular, fell from a leading position in the low-satisfaction model to the middle of the ranking under resampling. The relative prominence of these features should therefore not be overinterpreted.

Overall, the findings answer RQ1 and RQ3 by supporting compositional heterogeneity. The models shared several broadly consistent directional patterns, while customer tenure formed the clearest and most stable predictive core. Other predictors differed in their relative

prominence and combination across satisfaction segments. This is consistent with the argument that segmented modelling can reveal segment-specific drivers that a single aggregate analysis would not distinguish (De Caigny et al., 2021). However, the results describe how the fitted models used the observed predictors. They do not demonstrate that satisfaction caused these differences or that the identified features represent separate causal churn mechanisms.

### 5.2.2 Predictive Performance and Interpretive Boundaries

The pooled and satisfaction-specific models showed strong and consistent discrimination between customers recorded as churned and retained. Although the segment-specific models produced higher estimated scores than the pooled benchmark, this does not establish that satisfaction segmentation improved predictive accuracy, since each model was evaluated on a different analytical sample with its own composition and churn prevalence. Its main value was interpretive, as it allowed the relative importance of predictors to vary across customer groups.

The robustness checks reported in Section 4.4.3 did not identify a mechanical explanation for the unusually high performance: permutation reduced discrimination to chance level, and alternative preprocessing, feature exclusion and duplicate removal left ranking performance essentially unchanged. The separation therefore reflects predictive structure in the dataset rather than a signal confined to one feature or an evident pipeline artefact, although such checks cannot exclude every form of measurement or specification error.

As established in Section 3.3.1, the churn indicator registers a status assigned by the firm, and the point at which it was recorded is not documented. Where the status had already been assigned within the period the predictors summarise, the customer may have adjusted their behaviour accordingly, so part of the observed association may run from the status to the predictors rather than in the opposite direction. The two may also hold at once: if the satisfaction score and the churn status were recorded together following a single transaction, a customer could rate that purchase favourably while indicating that they would not purchase again. This reading accounts for two apparent contradictions in the results.

The first concerns the direction of days since last order. Its relationship with churn runs counter to what an inactivity-based definition would lead one to expect; but if the status was not derived from purchase intervals, that direction is not contradictory. The second concerns the direction of the satisfaction–churn relationship. Section 4.3.2 reported that recorded churn rises with satisfaction, contrary to the direction the literature would lead one to expect. Oliver (1999) treats satisfaction as necessary but insufficient for loyalty, allowing that satisfied customers may still defect. Churn increasing with satisfaction is a different claim. The two measures do not capture the same thing: satisfaction records an evaluation of completed transactions, and on the reading set out above both may hold simultaneously.

This is not treated as evidence of a general relationship between satisfaction and churn. Satisfaction was stratified on rather than modelled, so no effect was estimated; and the baseline differences between the three segments follow from the grouping rule, whose breakpoints were selected from this same distribution (Section 3.3.2). The rise across the original scale remains a descriptive property of this extract.

These two explanations concern how the associations should be interpreted; the undocumented point of recording also bears on the performance estimates themselves. RQ2 is therefore answered cautiously. Random cross-validation estimates performance within this dataset and cannot reproduce prospective prediction, which in churn research is established by separating a calibration period in which the predictors are observed from a later prediction period in which the outcome is recorded (Tamaddoni, Stakhovych and Ewing, 2016). The models demonstrated discrimination within this extract, not verified prospective accuracy; establishing the latter would require validation across another period, platform or customer population (Steyerberg and Harrell, 2016).

### 5.2.3 Integrating SHAP and RFM-Informed Behavioural Insights

SHAP and RFM provided complementary forms of interpretation. SHAP showed how predictors contributed within each satisfaction-specific model, whereas RFM converted selected behavioural variables into recognisable customer profiles, allowing profile-level patterns to be read within satisfaction-defined contexts.

Monetary value was the clearest point of convergence. Lower cashback amount pushed SHAP predictions towards churn, and lower monetary bands recorded higher churn rates in all three segments. The high-satisfaction segment nevertheless showed the widest difference between monetary bands even though cashback amount ranked lower in its SHAP model.

This is not contradictory. RFM compares observed outcomes across bands, whereas SHAP attributes contribution after the full predictor set has been considered (Lundberg et al., 2020), so a variable may show a clear profile-level association without adding the most information in a multivariate model. Frequency was the weakest dimension on its own, yet the highest-churn profiles combined limited order counts with the lower monetary bands, indicating that the dimensions identified these customers jointly.

The RFM findings also showed that recent activity was not sufficient evidence of retention security. Conventional RFM logic treats recent purchasing as a sign of stronger customer activity (Anitha and Patil, 2022), yet the most recent band recorded the highest churn rate in every segment. R3_F2_M1 further showed that recent purchasing could coexist with limited frequency and low monetary value, although the adjacent profile differing only in order count showed no comparable increase among highly satisfied customers.

The pattern therefore attaches to this profile rather than to low-monetary profiles as a class, and the unknown temporal ordering of the outcome prevents it from being read as evidence that recent purchasing increases churn; the profile identifies customers occupying a high-risk position, not a behaviour that produces one.

RQ4 is therefore answered by showing that RFM added an interpretable profile-level perspective, while satisfaction segmentation revealed that the same profile could carry different priorities across customer groups. SHAP provided segment-level context but could not establish the mechanism underlying R3_F2_M1, which should be treated as an exploratory pattern rather than a validated churn rule.

## 5.3 Managerial Implications and CRM Application

### 5.3.1 From Prediction to CRM Decision-Making

The findings can be translated into CRM decisions through a layered process. Because the outcome records an assigned status rather than confirmed attrition, the models identify customers who are disengaging rather than customers who have already left. This is the more actionable of the two for CRM purposes, since intervention is still possible. XGBoost first identifies customers whose characteristics resemble those of customers holding that position. This shows whom the firm may need to review, but not which action should follow. Predictive accuracy alone is therefore insufficient for managerial use (Chang et al., 2024).

SHAP adds explanation by showing which features carried most weight within each satisfaction segment's model. These features should be treated as diagnostic signals. Complaint status may prompt direct follow-up, while customer tenure, registered addresses and delivery distance may trigger reviews of relationship development, account complexity or delivery experience.

RFM profiles then present selected behaviours in a format that CRM teams can use more easily. Satisfaction segmentation provides further context because the same behavioural profile may carry different priorities across customer groups. This reflects the view that segmentation should generate actionable insight (Bailey et al., 2009) and that RFM can support targeted CRM decisions (Aslantaş et al., 2023).

### 5.3.2 Shared Retention Priorities

Across the three satisfaction segments, the clearest shared priority concerned the customer relationship stage rather than the digital activity indicators, none of which formed a consistently prominent predictive core. High application usage should therefore not be treated as evidence that a customer relationship is secure. Although digital touchpoint frameworks treat engagement as a route to loyalty (Nanta et al., 2025), high interaction may equally reflect transactional friction rather than commitment.

Customer tenure provided a stronger and more stable basis for CRM action. It was the most influential and stable feature across all three models, indicating that early-stage relationships warrant systematic attention regardless of satisfaction level. Short customer tenure could therefore serve as an early-lifecycle monitoring trigger, supporting relationship development through structured post-purchase onboarding, membership-benefit communication and timely engagement beyond the initial transaction, consistent with proactive retention logic (Ascarza et al., 2018). Complaint status could then serve as an escalation trigger: although the interaction between the two was not tested here, both were shared churn-related signals in the fitted models, so short-tenure customers with complaints could be prioritised for rapid resolution and follow-up (Section 5.3.3).

The RFM analysis identified R3_F2_M1 as a common high-churn profile, with a rate exceeding the corresponding baseline in all three satisfaction segments and reaching 64.0% among highly satisfied customers (Table 4.10). Recent activity was therefore not sufficient evidence of retention security, which also highlights a limitation of conventional heuristic RFM interpretations that treat recent purchasing and higher frequency as automatic indices of customer strength (Anitha and Patil, 2022).

Firms could flag recent activity combined with low monetary value within a customer's first few orders. The flag is a screening rule for review rather than a causal claim, and the timing is derived from the observed profiles rather than being a universal intervention point. As Section 4.6.3 notes, the adjacent one-order profile was also elevated, so the flag does not depend on a second purchase having been recorded. It should prompt a review of reward participation, complaint history and delivery experience rather than an automatic retention offer, since customers with the same profile may require different responses across satisfaction segments.

### 5.3.3 Segment-Differentiated Retention Strategies

Within the medium-satisfaction segment, complaint status emerged as a particularly prominent predictive signal. A moderate overall rating may therefore conceal a specific unresolved service failure. Morgeson III et al. (2020) show that the loyalty return on complaint recovery is positively moderated by satisfaction, so a uniform recovery standard will not pay

off equally across segments. Their result concerns the payoff of recovery; the present models indicate where complaint status is most diagnostic. The two need not coincide.

For medium-satisfaction customers with recorded complaints, CRM teams could establish a dedicated escalation pathway rather than route them through the standard process. Such a pathway might combine prompt resolution of the underlying failure with transparent handling procedures, empathetic communication and confirmation that the issue has been settled. The analysis identifies where complaint status is diagnostic, not how complaints are best handled.

Within the high-satisfaction segment, delivery distance and registered addresses became relatively more prominent. Neither should be read as a direct cause of churn. Both may instead indicate delivery-related or account-level complexity that an overall satisfaction score does not capture. Delivery service quality is an important component of the online retail experience (Vakulenko et al., 2019).

Two findings sharpen this concern. Inappropriate driver behaviour and inflexibility reduce customer satisfaction and repurchase intentions through anger and sadness respectively, and inappropriate behaviour harms outcomes regardless of whether the driver is outsourced (Masorgo, Mir and Hofer, 2023). Among customers who respond to a delivery problem by avoiding the retailer, 44% take no further action such as complaining or posting a review (Aljohani, 2024). Delivery dissatisfaction may therefore never appear in complaint records.

Firms should include last-mile delivery experience within the monitoring of highly satisfied customers, using measures such as delivery delays, failed deliveries, repeated address changes and tracking accuracy. Where fulfilment is outsourced, service-level agreements should cover driver conduct as well as delivery performance.

Within the low-satisfaction segment, lower cashback received emerged as a prominent churn-related signal. The other features ranking highly here, product category and city tier, were not stable under resampling and are therefore not developed into separate recommendations.

Lower cashback does not imply that firms should distribute indiscriminate discounts. Low monetary value is better used as a diagnostic and prioritisation signal. Firms should examine whether low-satisfaction customers understand and receive meaningful value from the

existing loyalty programme. Lower cashback may reflect limited participation, low purchasing activity, or a reward structure that is simply not relevant to them.

Clearer reward communication, more relevant loyalty benefits or progressively accumulated cashback are therefore worth considering for this group. Cashback accumulation and redemption both raise loyalty towards the programme, which in turn is associated with higher average purchase amounts and retailer revenues (Vieira et al., 2022). These remain strategic options rather than demonstrated effects.

These segment-specific recommendations should be treated as testable retention hypotheses. SHAP and RFM identify predictive associations; neither estimates whether additional rewards, revised recovery procedures or delivery improvements will causally reduce churn. This is consistent with the distinction between identifying customers at high risk and identifying those whose behaviour an intervention can actually change (Ascarza et al., 2018).

Firms should therefore pilot the proposed actions and compare their incremental effects separately within each satisfaction segment. Evaluation should consider repeat purchasing, complaint recurrence, incremental retention and intervention cost. Strategies should be scaled only when they generate commercially meaningful value.

# Chapter 6: Conclusions

## 6.1 Chapter Introduction

This chapter summarises how the research aim and the four research questions were addressed, then sets out the study's contributions, its main limitations and directions for future research.

## 6.2 Key Conclusions

Regarding RQ1, customers recorded as churned had shorter customer tenure, lower cashback amount and lower days since last order within every satisfaction segment, while transactional patterns such as order count and coupons used reversed direction across the scale. Recorded churn therefore showed a shared descriptive core rather than a uniform customer profile.

Regarding RQ2, because the churn indicator records a status assigned by the firm rather than an independently observed act of defection, and the point at which it was recorded is not documented, these results demonstrate separability within this dataset rather than verified prospective accuracy.

Regarding RQ3, customer tenure was the clearest and most stable shared predictor. The remaining differences concerned relative prominence: monetary value in the low-satisfaction model, complaint status in the medium-satisfaction model, and address- and distance-related features in the high-satisfaction model. The evidence supports compositional heterogeneity rather than entirely separate churn mechanisms.

Regarding RQ4, RFM converted selected behaviours into interpretable profiles, and satisfaction segmentation showed that the same profile could carry different priorities across customer groups. The elevated churn associated with R3_F2_M1 indicated that recent activity was not sufficient evidence of retention security, although the profile remains exploratory. RFM therefore supports operational profiling, SHAP provides diagnostic context, and satisfaction informs the prioritisation of CRM responses.

## 6.3 Contributions and Managerial Relevance

The principal analytical contribution lies in treating satisfaction as a stratifying context rather than as one predictor within an aggregate model. Segmentation on this basis revealed a shared predictive core alongside differences in the relative prominence of other customer characteristics, providing evidence of compositional heterogeneity.

The study also demonstrates the complementary value of prediction, explanation and behavioural profiling. XGBoost examined complex patterns among multiple predictors without requiring all relationships to be specified in advance, SHAP showed how the fitted models used those predictors, and RFM converted selected behaviours into recognisable customer profiles. Together they support a CRM process in which firms identify customers for review, investigate the relevant signals and tailor the response to the customer's satisfaction context. The contribution is therefore a framework for replacing undifferentiated retention activity with more focused and testable decisions, rather than a universal retention rule.

## 6.4 Limitations and Future Research

Most importantly, the outcome variable records a status rather than an observed act of defection, and neither the procedure by which it was assigned nor the point at which it was recorded is documented. The analysis therefore identifies characteristics associated with that status rather than antecedents of subsequent behaviour. Future research should use longitudinal customer histories with defined observation and prediction windows, so that predictors are observed in advance of the outcome and churn can be operationalised as a business-appropriate period of purchase inactivity.

Two measures were also constrained. Satisfaction was recorded as a single overall score that does not distinguish product, service recovery, digital interaction or delivery experience, and the RFM dimensions relied on proxies, with cashback standing in for the expenditure the dataset does not record. Future studies could apply multidimensional satisfaction measures and actual transaction values, and compare satisfaction-based groups with segmentations based on lifecycle stage, customer value or data-driven clustering.

Finally, the findings derive from a single e-commerce dataset and describe association rather than effect. They require validation across other periods, platforms and customer populations, and the proposed retention strategies should be tested through controlled field experiments to determine whether changes to rewards, complaint handling or delivery processes generate incremental retention and commercially meaningful value.

## 6.5 Final Statement

Satisfaction alone was insufficient to characterise recorded churn. Customers across satisfaction segments shared a stable predictive core, but the relative prominence of other signals differed. Retention strategy should therefore combine common lifecycle priorities with segment-sensitive diagnosis, while treating predictive insights as hypotheses that require prospective validation.

# References


Ahmad, A.K., Jafar, A. and Aljoumaa, K., 2019. Customer churn prediction in telecom using machine learning in big data platform. *Journal of big data,* 6(1), pp.1-24.

Al-Shatnwai, A.M. and Faris, M., 2020. Predicting customer retention using XGBoost and balancing methods. *International journal of advanced computer science and applications,* 11(7).

Aljohani, K., 2024. The role of last-mile delivery quality and satisfaction in online retail experience: An empirical analysis. *Sustainability,* 16(11), p.4743.

Anitha, P. and Patil, M.M., 2022. RFM model for customer purchase behavior using K-Means algorithm. *Journal of King Saud University - computer and information sciences,* 34(5), pp.1785-1792.

Ascarza, E., Neslin, S.A., Netzer, O., Anderson, Z., Fader, P.S., Gupta, S., Hardie, B.G., Lemmens, A., Libai, B. and Neal, D., 2018. In pursuit of enhanced customer retention management: Review, key issues, and future directions. *Customer needs and solutions,* 5(1), pp.65-81.

Aslantaş, G., Gençgül, M., Rumelli, M., Özsaraç, M. and Bakırlı, G., 2023. Customer segmentation using K-means clustering algorithm and RFM model. *Dokuz Eylül Üniversitesi mühendislik fakültesi fen ve mühendislik dergisi,* 25(74), pp.491-503.

Athanassopoulos, A.D., 2000. Customer satisfaction cues to support market segmentation and explain switching behavior. *Journal of business research,* 47(3), pp.191-207.

Bailey, C., Baines, P.R., Wilson, H. and Clark, M., 2009. Segmentation and customer insight in contemporary services marketing practice: why grouping customers is no longer enough. *Journal of marketing management,* 25(3-4), pp.227-252.

Boukrouh, I. and Azmani, A., 2025. Explainable machine learning models applied to predicting customer churn for e-commerce. *IAES international journal of artificial intelligence,* 14(1), pp.286-297.

Brahmana, R.W.S., Mohammed, F.A. and Chairuang, K., 2020. Customer segmentation based on RFM model using K-means, K-medoids, and DBSCAN methods. *Lontar komputer: jurnal ilmiah teknologi informasi,* 11(1), p.32.

Chang, V., Hall, K., Xu, Q.A., Amao, F.O., Ganatra, M.A. and Benson, V., 2024. Prediction of customer churn behavior in the telecommunication industry using machine learning models. *Algorithms,* 17(6), p.231.

Chen, T. and Guestrin, C., 2016. Xgboost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, 13-17 August 2016, San Francisco, Calif. New York: ACM, pp.785-794.

Cooil, B., Aksoy, L. and Keiningham, T.L., 2007. Approaches to customer segmentation. *Journal of relationship marketing,* 6(3-4), pp.9-39.

De Caigny, A., Coussement, K., Verbeke, W., Idbenjra, K. and Phan, M., 2021. Uplift modeling and its implications for B2B customer churn prediction: A segmentation-based modeling approach. *Industrial marketing management,* 99, pp.28-39.

Dick, A.S. and Basu, K., 1994. Customer loyalty: toward an integrated conceptual framework. *Journal of the academy of marketing science,* 22(2), pp.99-113.

Fuentes-Blasco, M., Moliner-Velázquez, B. and Gil-Saura, I., 2014. Effect of customer heterogeneity on the relationship satisfaction–loyalty. *Revista española de investigación de marketing ESIC,* 18(2), pp.78-92.

Füller, J. and Matzler, K., 2008. Customer delight and market segmentation: An application of the three-factor theory of customer satisfaction on life style groups. *Tourism management,* 29(1), pp.116-126.
Gan, L., 2022. XGBoost-Based E-Commerce Customer Loss Prediction. *Computational intelligence and neuroscience,* 2022(1), p.1858300.
Grewal, R., Chandrashekaran, M. and Citrin, A.V., 2010. Customer satisfaction heterogeneity and shareholder value. *Journal of marketing research,* 47(4), pp.612-626.
Human, G. and Naude, P., 2014. Heterogeneity in the quality–satisfaction–loyalty framework. *Industrial marketing management,* 43(6), pp.920-928.
Jahan, I. and Sanam, T.F., 2026. A comprehensive framework for customer retention in E-commerce using machine learning based on churn prediction, customer segmentation, and recommendation. *Electronic commerce research,* 26(1), pp.1-44.
Kim, S., Fong, D.K. and DeSarbo, W.S., 2012. Model-based segmentation featuring simultaneous segment-level variable selection. *Journal of marketing research,* 49(5), pp.725-736.
Kohavi, R., 1995. A study of cross-validation and bootstrap for accuracy estimation and model selection. *Proceedings of the 14th international joint conference on artificial intelligence (IJCAI-95)*, 20-25 August 1995, Montreal. Morgan Kaufmann, pp.1137-1143.
Kumar, V., Dalla Pozza, I. and Ganesh, J., 2013. Revisiting the satisfaction–loyalty relationship: empirical generalizations and directions for future research. *Journal of retailing,* 89(3), pp.246-262.
Lundberg, S.M., Erion, G., Chen, H., DeGrave, A., Prutkin, J.M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N. and Lee, S.-I., 2020. From local explanations to global understanding with explainable AI for trees. *Nature machine intelligence,* 2(1), pp.56-67.
Lundberg, S.M. and Lee, S.-I., 2017. A unified approach to interpreting model predictions. *Advances in neural information processing systems*, 4-9 December 2017, Long Beach, Calif., pp.4765-4774.
Masorgo, N., Mir, S. and Hofer, A.R., 2023. You're driving me crazy! How emotions elicited by negative driver behaviors impact customer outcomes in last mile delivery. *Journal of business logistics,* 44(4), pp.666-692.
Matuszelański, K. and Kopczewska, K., 2022. Customer churn in retail e-commerce business: Spatial and machine learning approach. *Journal of theoretical and applied electronic commerce research,* 17(1), pp.165-198.
Maulana, B.A. and Hidayati, N., 2025. Churn Prediction in Credit Customers Using Random Forest and XGBoost Methods. *Indonesian journal of data and science,* 6(1), pp.81-89.
Morgeson III, F.V., Hult, G.T.M., Mithas, S., Keiningham, T. and Fornell, C., 2020. Turning complaining customers into loyal customers: Moderators of the complaint handling–Customer loyalty relationship. *Journal of marketing,* 84(5), pp.79-99.
Morwitz, V.G., Steckel, J.H. and Gupta, A., 2007. When do purchase intentions predict sales? *International journal of forecasting,* 23(3), pp.347-364.
Nanta, T.M., Noermijati, N., Rohman, F. and Hussein, A.S., 2025. The effect of digital touchpoint usage experience on customer loyalty mediated by digital engagement and customer satisfaction. *Businesses,* 5(1), p.3.
Oliver, R.L., 1999. Whence consumer loyalty? *Journal of marketing,* 63(4_suppl1), pp.33-44.
Ringle, C.M., Sarstedt, M. and Mooi, E.A., 2010. Response-based segmentation using finite mixture partial least squares: theoretical foundations and an application to American

customer satisfaction index data. In: R. Stahlbock, S.F. Crone and S. Lessmann, eds. *Data mining: special issue in annals of information systems.* New York: Springer, pp.19-49.
Steyerberg, E.W. and Harrell, F.E., 2016. Prediction models need appropriate internal, internal-external, and external validation. *Journal of clinical epidemiology,* 69, p.245.
Tamaddoni, A., Stakhovych, S. and Ewing, M., 2016. Comparing churn prediction techniques and assessing their performance: A contingent perspective. *Journal of service research,* 19(2), pp.123-141.
Ullah, A., Mohmand, M.I., Hussain, H., Johar, S., Khan, I., Ahmad, S., Mahmoud, H.A. and Huda, S., 2023. Customer analysis using machine learning-based classification algorithms for effective segmentation using recency, frequency, monetary, and time. *Sensors,* 23(6), p.3180.
Vakulenko, Y., Shams, P., Hellström, D. and Hjort, K., 2019. Online retail experience and customer satisfaction: the mediating role of last mile delivery. *The international review of retail, distribution and consumer research,* 29(3), pp.306-320.
Verma, A., 2021. *Ecommerce customer churn analysis and prediction* [Online]. Kaggle. Available from: https://www.kaggle.com/datasets/ankitverma2010/ecommerce-customer-churn-analysis-and-prediction [Accessed 2 June 2026].
Vieira, V.A., Agnihotri, R., De Almeida, M.I.S. and Lopes, E.L., 2022. How cashback strategies yield financial benefits for retailers: The mediating role of consumers' program loyalty. *Journal of business research,* 141, pp.200-212.
Wilbert, H.J., Hoppe, A.F., Sartori, A., Stefenon, S.F. and Silva, L.A., 2023. Recency, frequency, monetary value, clustering, and internal and external indices for customer segmentation from retail data. *Algorithms,* 16(9), p.396.
Zaghloul, M., Barakat, S. and Rezk, A., 2025. Enhancing customer retention in Online Retail through churn prediction: A hybrid RFM, K-means, and deep neural network approach. *Expert systems with applications,* 290, p.128465.